Check for updates

DOI: 10.1002/geot.202300007

Chee-Min Khoo, Teik-Aun Ooi

Geotechnical challenges and innovations in urban underground construction – The Klang Valley Mass Rapid Transit project

The Klang Valley Mass Rapid Transit (KVMRT) project involves the construction of an urban passenger transport system, i.e., Mass Rapid Transit (MRT) system, together with the existing urban rail network, will form the backbone of the public transport system in the Greater Kuala Lumpur/Klang Valley region in Malaysia. The first MRT line implemented is the 47 km Kajang Line, of which 9.5 km is underground tunnels with seven underground stations. Construction of the line began on 8 July 2011 and achieved the full line opening on 17 July 2017. The second MRT line—Putrajaya Line—began fully operations on 16 March 2023, stretches from Sungai Buloh to Serdang and ends at Putrajaya for a length of 57.7 km, of which 13.5 km is underground tunnels with 10 new underground stations. Like all other metro underground stations/tunnels designed as part of the urban city, the KVMRT underground construction faces challenges from a technical (engineering design and construction) as well as social (environmental and land related) point of view. Needless to say, underground construction in Klang Valley is intensified with the inherent geotechnical challenges presented by complex ground conditions ranging from hard granite, heterogeneous Kenny Hill formation and extreme karstic limestone with fully developed weathered profiles to soft recent deposits including alluvium and mine tailing that is under-consolidated in places due to past mining activities. The development of this large-scale infrastructure project has not only opened up tremendous works and new frontier for tunnelling and geotechnical engineering in Malaysia, but it also provided a wealth of information of unique geotechnical challenges/accomplishments as well as technology breakthrough and design innovation in underground engineering. This article discusses some aspects of geotechnical challenges and interesting lessons learnt as well as numerous innovations embedded into this highly complex urban underground construction project.

Keywords tunnelling technology; complex geology; ground improvement; instrumentation; innovations

1 Introduction

1.1 The Klang Valley Mass Rapid Transit project

The Klang Valley Mass Rapid Transit (KVMRT) project involves the construction of an urban passenger transport system, i.e., Mass Rapid Transit (MRT) system, together with the existing urban rail network, will form the backbone of the public transport system in the Greater Kuala Lumpur/Klang Valley region in Malaysia. The project consists of three metro lines envisaging a 'wheel and spoke' concept with two northwest–southeast radial lines and one circle line looping around Kuala Lumpur. It aims to radically improve and transform Kuala Lumpur's poor and sorely inadequate public transportation coverage and to propel the Greater Kuala Lumpur metropolitan area to be on par with that of a developed city. The proposal was announced in June 2010 and was approved by the Government of Malaysia in December 2010.

The first MRT line implemented is the 47 km Kajang Line, of which 9.5 km runs underground beneath the centre of Kuala Lumpur while the rest of the alignment is elevated. The line has 29 stations, of which seven are underground. Construction of the MRT Line 1 began on 8 July 2011 and achieved the full line opening on 17 July 2017. The Putrajaya Line is the second MRT line set for full opening on 16 March 2023, which stretches from Sungai Buloh to Serdang and ends at Putrajaya. The alignment will have a length of 57.7 km, consisting of 44.2 km of elevated tracks and 13.5 km running through underground tunnels. A total of 38 stations, 11 of them underground, were built. Line 2 provides interchange with Line 1 at the existing TRX Station. The circle line (MRT Line 3), which is currently under reference design stage, will bring greater connectivity in Kuala Lumpur as it loops around the business districts of the city and serves an important role to tie up and integrate the currently disjointed LRT, KTM, and monorail lines. Given the terrain of the city and the number of existing developments, it is likely that a large proportion of the circle line is to be underground [1]. Figure 1 shows the alignment maps of KVMRT Kajang Line (Line 1) and Putrajaya Line (Line 2).

The KVMRT project has not only opened up tremendous works and new frontier for tunnelling and geotechnical engineering in Malaysia, but it also provided a wealth of information of unique geotechnical challenges/accomplishments as well as technology breakthrough and design innovation in underground engineering. Needless to say, underground construction and tunnelling in urban or build-up environment is very challenging and entails high risks [2]. Most of the geotechnical challenges and risks could be mitigated/managed through design, construction planning, and execution.

1.2 Underground works

Like all other underground metro designed as part of the metropolis, the need for underground space for the

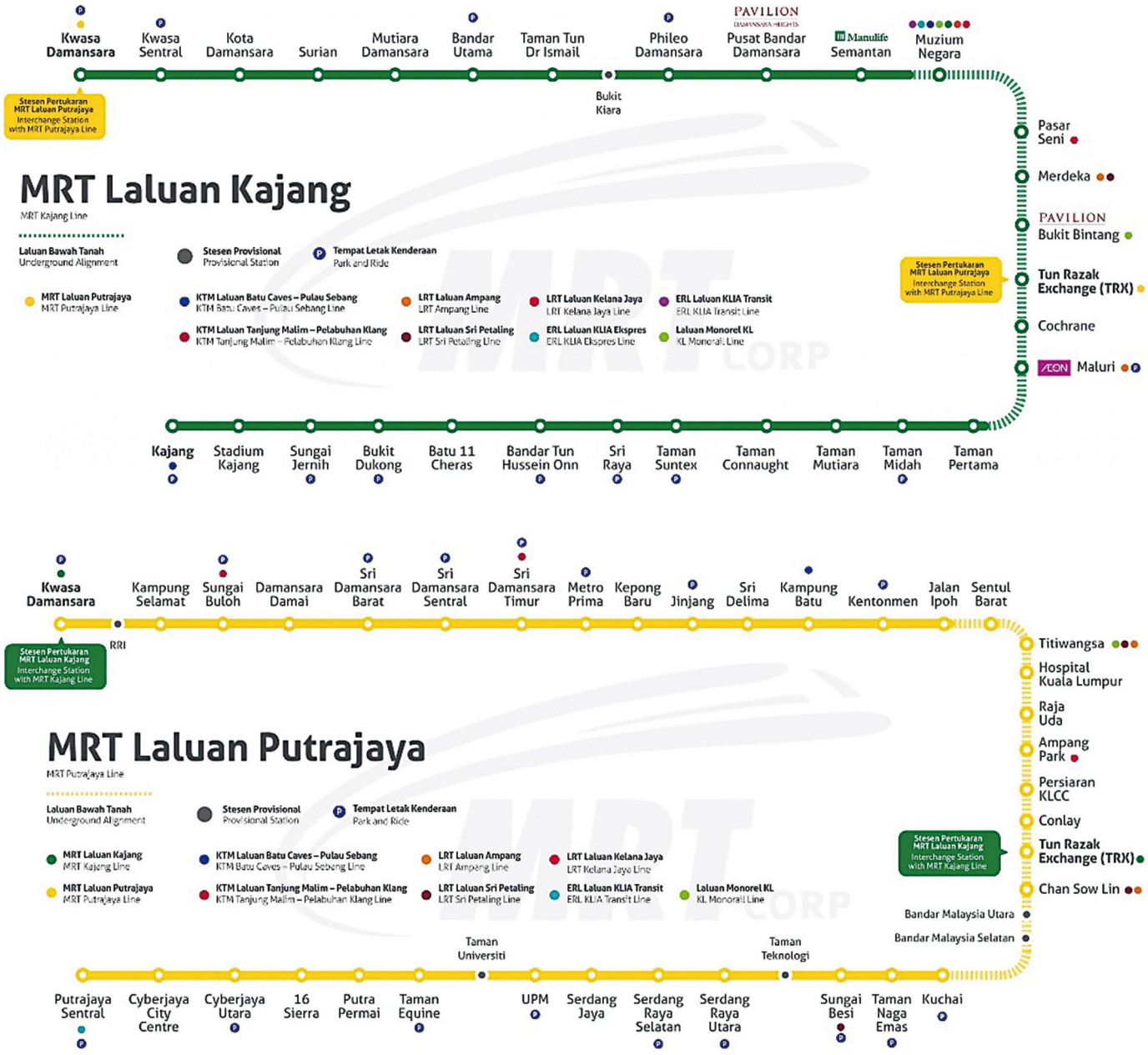


Fig. 1 Alignment maps of KVMRT Kajang Line (Line 1) and Putrajaya Line (Line 2)

KVMRT development necessitated deeper and yet complex underground construction works in the densely populated urban area of Kuala Lumpur. The underground works of the KVMRT essentially comprise the construction and completion of metro stations, cut-and-cover crossover boxes, twin bored tunnels, and its auxiliary structures such as cross passengers, escape shafts, intervention ventilation shafts, and portals. Figure 2 exemplifies the underground stations and online/offline shafts with different typologies and construction methodologies.

Typically, an underground station box measures about 200 m long and 22 m wide in either island platform or stacked platform. The depths of stations could be varying significantly mainly depending on the alignment requirement. It is preferably for better commuter experience that metro stations are not situated deep underground; however, the need to go deeper and deeper for certain stations and tunnel sections is observed in Line 2 as the top layers are already congested with various building structures and infrastructures, not to mention the already congested site environment. In some cases, thick diaphragm wall of up to 1.5 m was constructed for the stacked station at great depth as compared to the common thickness of diaphragm wall ranges from 0.6 to 1.0 m used for basement construction in Kuala Lumpur. Toe depths of diaphragm walls are in the range of 30 to over 70 m depths depending on the final elevation of the excavation. Tab. 1 provides informative details of underground stations/shafts for both the Line 1 and Line 2. Figure 3 illustrates the embedded length and thickness of diaphragm wall in relation to the excavation depths for each station/shaft.

It is to be noted that for station excavation in limestone formation, anchored secant bored pile (SBP) wall or shallow deep soil mixing (DSM) gravity wall is used to retain the soil cover and followed by vertical excavation into limestone bedrock without the need for temporary sup-

18657389, 2023, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

18637849, 2023, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

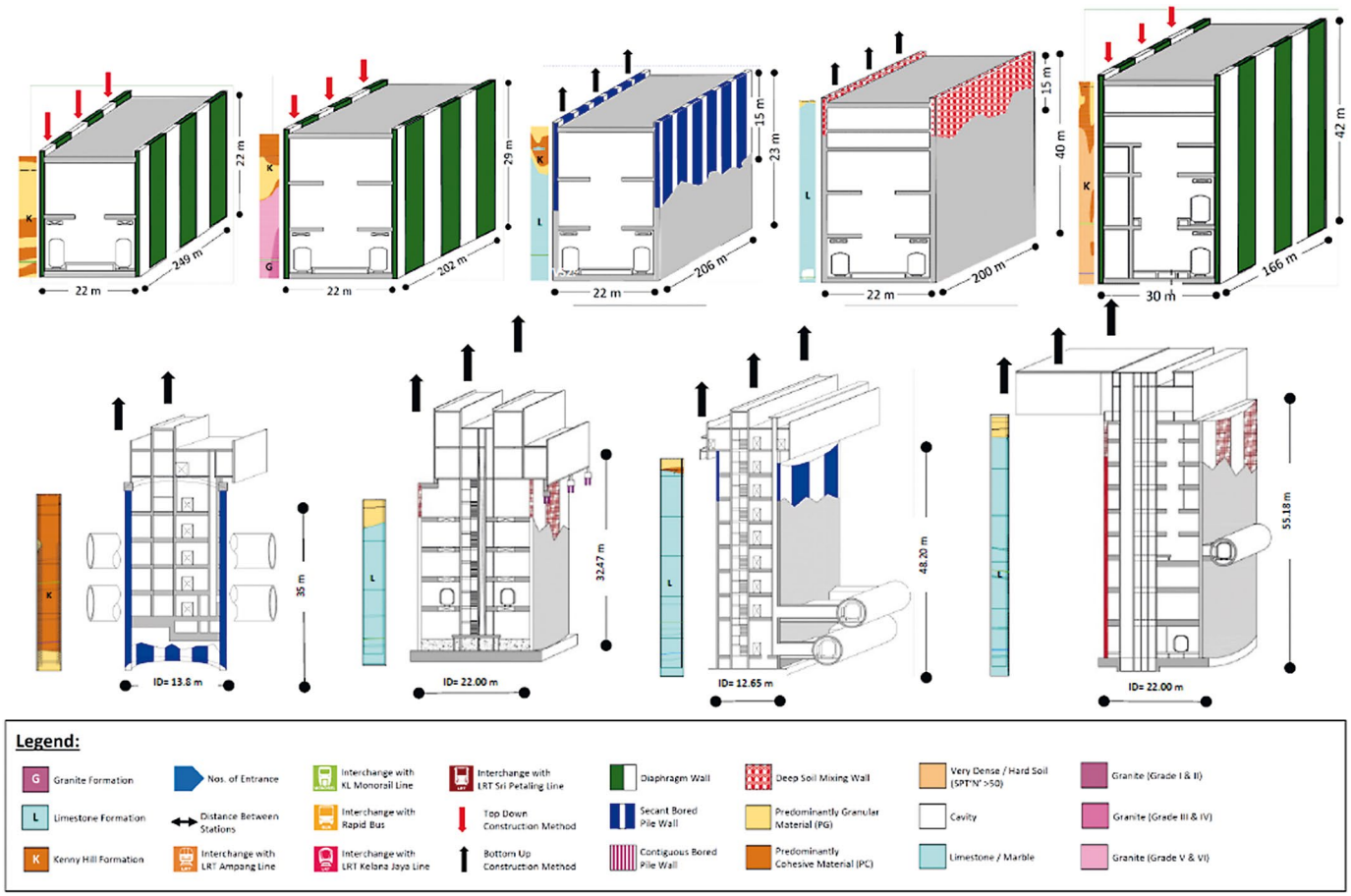


Fig. 2 Examples of underground stations and online/offline shafts

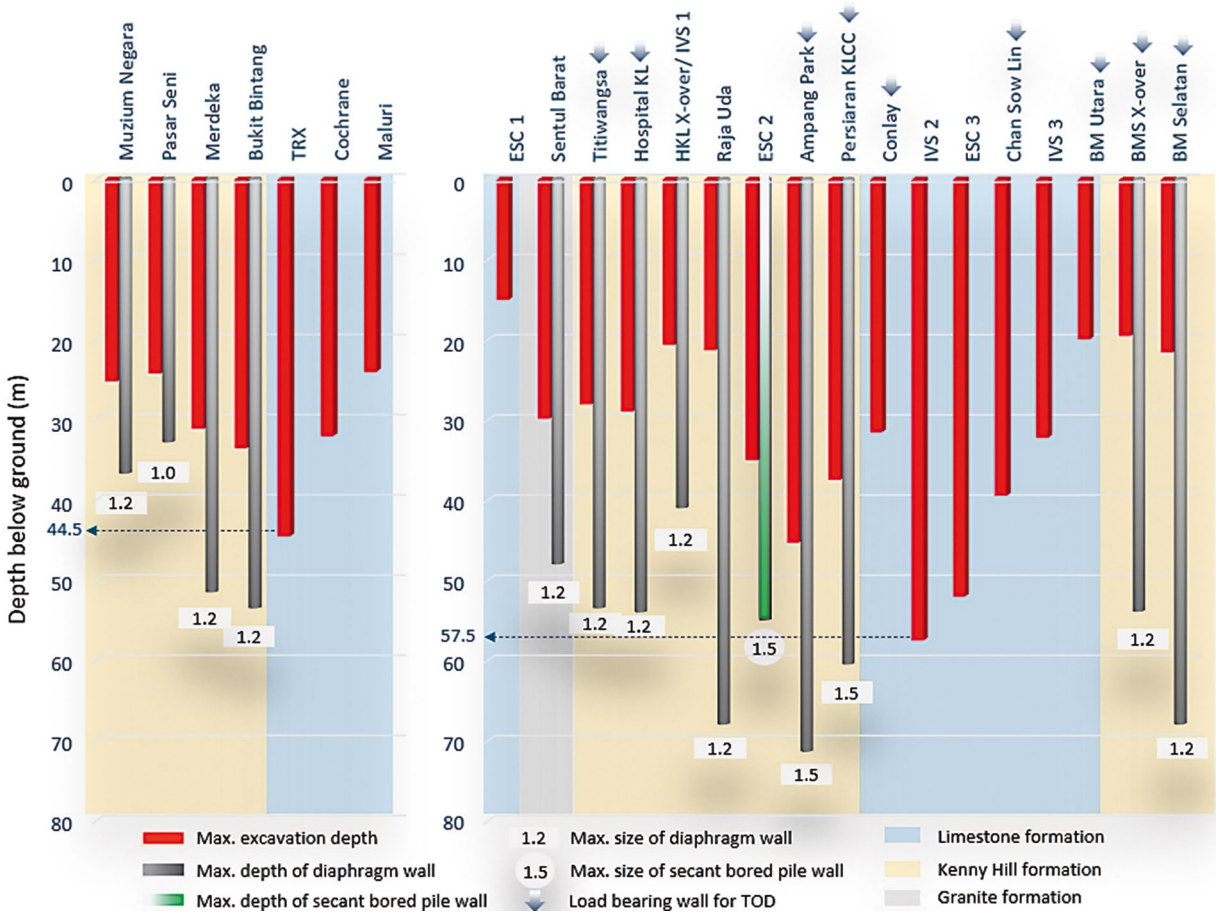


Fig. 3 Embedded length and thickness of diaphragm wall in relation to the excavation depths for each station/shaft

Tab. 1 Key information of underground stations and shafts

Line	Station/shaft	Type of station/ shaft	Anticipated geology	General dimension	Depth of excavation	Retention system
1	Muzium Negara	Island platform	Kenny Hill	166 m × 24 m	25.5 m	Diaphragm wall
1	Pasar Seni	Island platform	Kenny Hill	161.3 m × 24 m	25.1 m	Diaphragm wall
1	Merdeka	Island platform	Kenny Hill	148 m × 24 m	30.4 m	Diaphragm wall
1	Bukit Bintang	Stacked platform	Kenny Hill	138 m × 23 m	33.4 m	Diaphragm wall
1	TRX	Stacked platform	Limestone	168 m × 27.6 m	44.5 m	SBP wall
1	Cochrane	Island platform	Limestone	176 m × 21.3 m	32.0 m	SBP wall
1	Maluri Crossover	–	Limestone	125 m × 26.3 m	21.0 m	SBP wall
1	Maluri	Island platform	Limestone	153 m × 21.3 m	21.0 m	SBP wall
2	ESC 1	Online shaft	Limestone	33.3 m × 21.1 m	15.0 m	SBP wall
2	Sentul Barat	Island platform	Granite	202 m × 21.9 m	29.8 m	Diaphragm wall
2	Titiwangsa	Island platform	Granite/Kenny Hill	215 m × 26 m	27.0 m	SBP/Diaphragm wall
2	Hospital KL	Island platform	Kenny Hill	139 m × varies	28.9 m	Diaphragm wall
2	HKL Crossover/IVS 1	Online shaft	Alluvium/Limestone	226 m × 30.7 m	20.6 m	Diaphragm wall
2	Raja Uda	Island platform	Alluvium/Limestone	255 m × 21.8 m	21.3 m	Diaphragm wall
2	ESC 2	Offline shaft	Alluvium/Limestone	13.3 m diameter	35.0 m	SBP wall
2	Ampang Park	Stacked platform	Kenny Hill	162.5 m × 26 m	44.4 m	Diaphragm wall
2	Persiaran KLCC	Stacked platform	Kenny Hill	179 m × 23.0 m	37.5 m	Diaphragm wall
2	Conlay	Island platform	Limestone	174 m × 22.2 m	30.5 m	SBP wall
2	IVS 2	Offline shaft	Limestone	20.7 m diameter	57.5 m	DSM wall
2	ESC 3	On-/Offline shaft	Limestone	15.2 m diameter	52.0 m	SBP wall
2	Chan Sow Lin	Island platform	Limestone	224 m × 21.6 m	38.0 m	DSM/SBP wall
2	IVS 3	Online shaft	Limestone	20.9 m diameter	32.2 m	SBP wall
2	Bandar Malaysia Utara	Island platform	Limestone	255 m × 21.6 m	18.0 m	SBP wall
2	Bandar Malaysia Selatan Crossover	–	Limestone	190 m × 25 m	19.3 m	Diaphragm wall
2	Bandar Malaysia Selatan	Island platform	Limestone/Kenny Hill	294 m × 24 m	21.2 m	Diaphragm wall

Notes: ESC, escape shaft; IVS, intervention ventilation shaft.

port structures, except provision of curtain grouting and rock stabilization measures (see Figure 4). Tan et al. [3] presented the unique experience and design principles of temporary SBP wall retention systems in soil together with further vertical excavation in rock to depths ranging from 20 to more than 35 m below the toe of the SBP piles for station excavation in karstic limestone formation. The use of DSM wall to retain the overburden soil in Line 1



Fig. 4 Examples of soil retention and vertical excavation in limestone formation: a) deep soil mixing wall in Line 1 Maluri South Portal; b) anchored secant bored pile wall in Line 2 Chan Sow Lin Station

18657389, 2023, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

1865781, 2023, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Maluri South Portal was reported by Yee and Tan [4]. A shaft is commonly designed to be in circular shape to allow for strut-free excavation by harnessing its ability to transfer the lateral forces in the form of hoop stresses. The 57.5 m deep excavation carried out in IVS2 of Line 2 sets the deepest excavation ever done in Kuala Lumpur. Tan et al. [5] discussed the design and construction challenges for this deep shaft excavation.

Meantime, the tunnels run under the downtown area, where there are many high-rise buildings, roads, and bridges as well as many underground utilities, and also undercrossing rivers. The twin bored tunnels have an internal diameter of 5.8 m. The tunnels are constructed using 1.4 m wide, 275 mm-thick precast steel fibre reinforced concrete (SFRC) lining segments comprising seven segments plus one key. For most parts along the tunnel route, the twin tunnels were at a parallel side-by-side configuration and located at the same elevation except where it approaches the stacked stations. The successful completion of KVMRT Line 1 tunnels marks an important milestone in the use of SFRC tunnel segmental lining in South-East Asia [6]. Poh et al. [7] and Lim and Ng [8] gave the brief development of SFRC tunnel lining in Malaysia as well as the local challenges encountered and lessons learned during its initial design development and production phrases.

2 Surface and underground conditions

2.1 Topography and setting

Basically, the underground alignment passes through a highly populated area of Kuala Lumpur city with high-rise buildings, roads, and bridges and even rivers, namely, Sungai Kelang, Sungai Gombak, and Sungai Bunus. Underground construction in Kuala Lumpur is not only challenging due to the inherent complex geology but intensified with made ground associated with history of past tin mining activities in Kuala Lumpur since 1850s. The mining activities were widespread within the parts of the city and typically took place in large and deep open excavation pits in the areas underlain by the Kuala Lumpur Limestone. As shown in Figure 5, the pits are known to have been backfilled with mine tailing that is underconsolidated in places during subsequent mining activities or backfilled or remediated sometime after mining to allow for developments. This difficult ground condition proves to be challenging for underground construction as encountered in some areas of Line 2 alignment.

2.2 Geology and hydrogeology

The geological condition of Kuala Lumpur area is complex. Generally, there are four metasedimentary rock formations occurred in Kuala Lumpur, followed by the emplacement of the granite and the deposition of recent alluvium. The oldest of these geological units is the

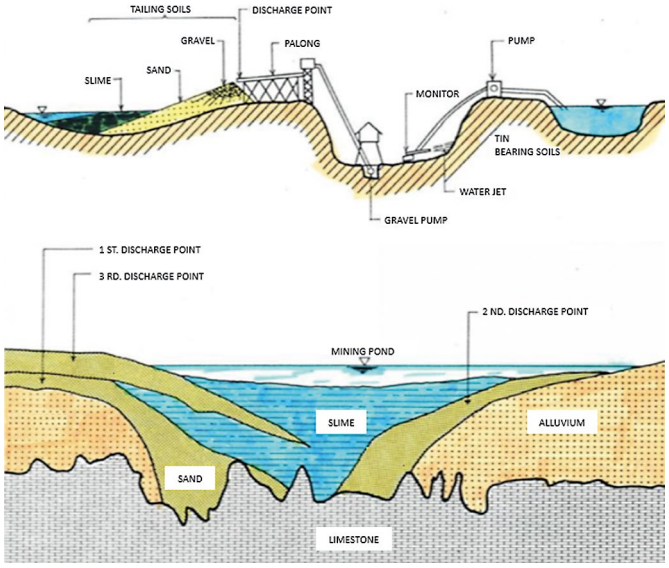


Fig. 5 Formation of tin mine tailing using gravel pump method

Hawthornden formation, consisting thick sequence of schists and phyllite overlain by the Kuala Lumpur Limestone which has yielded fossils of Middle to Upper Silurian age. The youngest known metasedimentary rocks in Kuala Lumpur area consist of an alternating succession of quartzite and phyllite, named the Kenny Hill formation. The intrusion of granite rock has resulted in the deformation of the older rocks; granite rock occupies a large hilly to undulating area around Kuala Lumpur. Alluvium is collectively the deposits laid down by the rivers. It forms an extensive surficial soil layer in the Klang Valley mainly along the major rivers such as the Sungai Kelang and Sungai Gombak. Alluvium is underlain by other soils such as the Kenny Hill formation, with the thickness of alluvium generally less than 10 m but can be thicker when it fills the trough between limestone pinnacles.

Geologically, the Line 1 bored tunnels run through two distinctive geological formations, namely, the Kuala Lumpur Limestone and Kenny Hill formation. However, the construction of Line 2 will pass through all the major geological formations in Kuala Lumpur–Kenny Hill formation, Kuala Lumpur Limestone, granite, and alluvium. Both alignments are passing through few major faulting lines. However, the faults are all inactive faults and there are no evidence or past occurrences of reactivation of these faults. Nevertheless, the fault zone area is usually at the contact zone between two different formations; therefore, the rock mass at these areas is more complicated and relatively weak. Quartz reef is located close to the Line 2 alignment at Hospital Kuala Lumpur Station. Figure 6 illustrates the anticipated geology along the tunnel alignments while Tab. 1 can be referred to for the station geology.

The hydrogeology of Kuala Lumpur is equally complex. Generally, the Kenny Hill formation, Kuala Lumpur Limestone, the residual soils of granite, and the alluvium deposits are permeable and thus high groundwater poten-

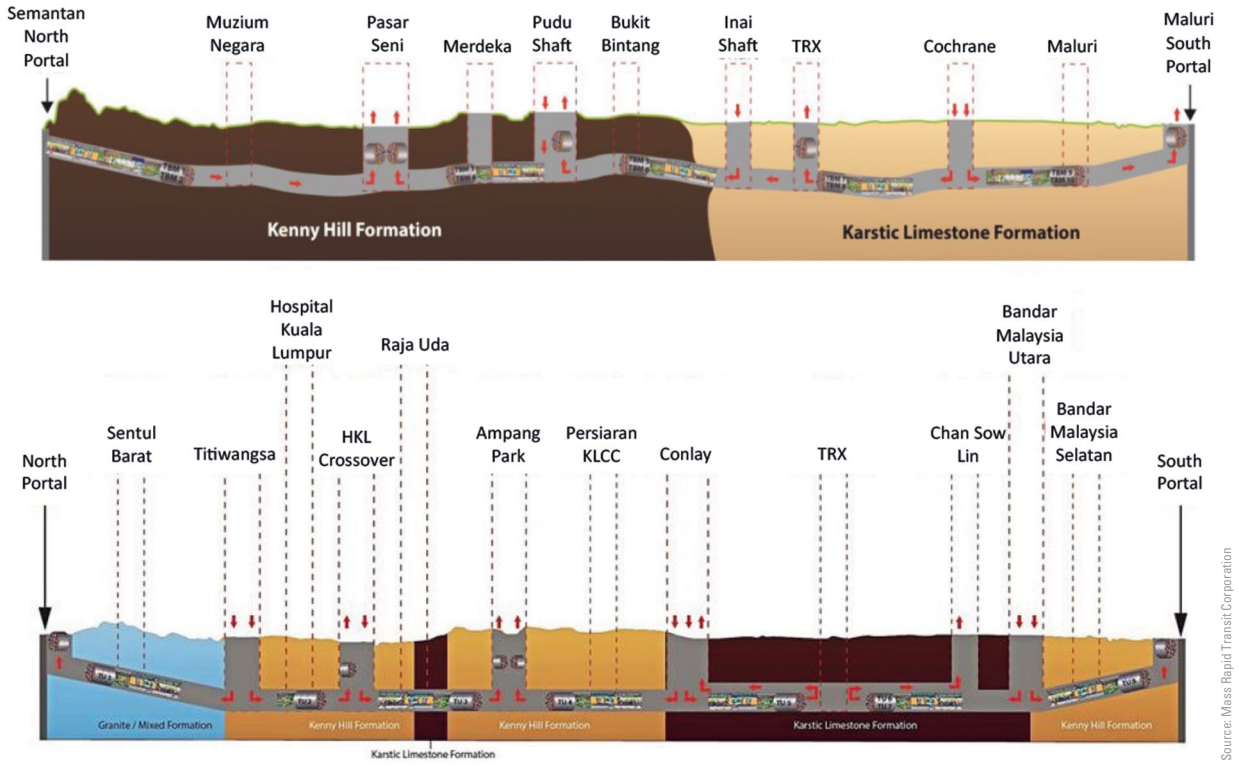


Fig. 6 Anticipated geology along the tunnel alignments

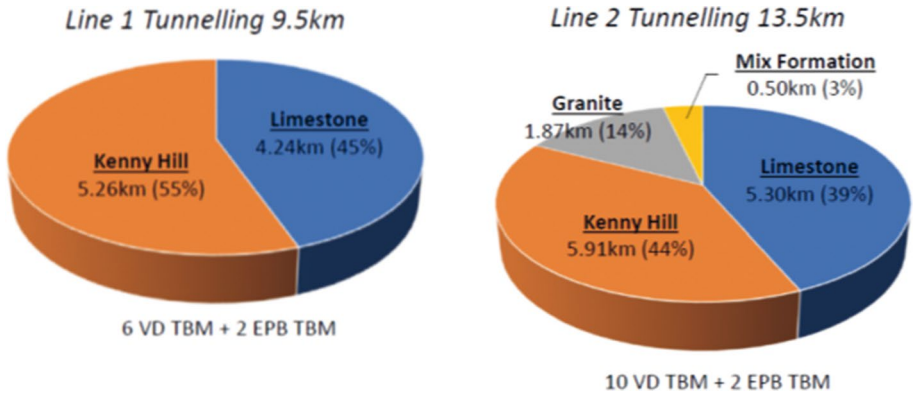


Fig. 7 Distribution of geological formations for tunnelling

tial. In particular, the limestone contains many fractures; solution channels and cavities created by chemical erosion of the carbonate may have relatively high hydraulic conductivity [9]. Because of the high permeability of the limestone, the ex-tin mining ponds play an important role in regulating Kuala Lumpur's groundwater regime, providing water storage during heavy rains and being a major source of groundwater recharge. The groundwater table in Kuala Lumpur is typically quite high because of the natural valley terrain and generally follows the ground topography and close to the surface.

2.2.1 The Kuala Lumpur Limestone

It is to be noted that almost half of the underground alignments are within the limestone (see pie charts in Figure 7). The Kuala Lumpur Limestone is well known for its erratic karstic features such as the pinnacle zone (i.e., highly irregular bedrock profile or rockhead, with pinnacles alternating with deep troughs), cavities and solution channels, overhangs, floaters/boulders, collapsed weak soil zone, or slump materials just above bedrock with SPT value equals to 0. The Kuala Lumpur Limestone is widely known among the local engineering fraternity due mainly to its myriad or slew of problems that can arise in the construction works in the limestone (piling, excavation, tunnelling, etc.). Construction problems associated with limestone have been the major theme of conferences held in Kuala Lumpur in the past (e.g., the Southeast Asian Geotechnical Conference in 1985, 1996, 2007, 2016; the IEM International Tunnelling Conference in 2006, 2011, 2015, 2017, etc.).

Tunnelling through the pinnacle zone would mean having to contend with mixed face of limestone rock and soft soils (alluvium) and/or full face of soft soils. The notorious problems are further exacerbated by the fact that the

18657389, 2023, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TOPICS Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

limestone bedrock is often overlain by weak alluvium which, combined with the nature of the limestone, becomes highly susceptible or prone to excessive ground settlement if the face pressure is not properly controlled during tunnelling. Klados et al. [10] highlighted the challenges of tunnelling through Kuala Lumpur karst formation. They presented the actual performance of the variable density tunnel boring machine (TBM) through case studies of the operation data recorded along the tunnel drives completed in Line 1 tunnelling. The tunnelling works in Kuala Lumpur Limestone were successfully completed with variable density TBM with significantly fewer incidents compared to the SMART project (i.e., 41 numbers of major ground collapses); this translates to 4.6/km sinkholes for SMART versus 0.3/km sinkholes for KVMRT Line 1 [11].

The unpredictability of rock levels in notorious karst limestone with high groundwater table adds on to the list of challenges for the station/shaft excavation. Goh et al. [12] and Tan et al. [13] outlined the strategies for successful management and mitigation of the challenges of deep excavation in limestone formation with the case study of Line 1 Maluri Station located in highly urbanized area. Tight regulatory requirements for rock blasting were extremely challenging given the close proximity of pre-existing and temporary structures. Controlled rock blasting was successfully carried out up to 1.0 m away from king posts supporting traffic decking at Maluri Station. This expedites the construction progress with minimal traffic disruption. The challenges of deep excavation work in limestone formation also bear some interesting aspects which will be discussed in the subsequent sections.

2.2.2 The Kenny Hill formation

An equally unique and significant geology encountered in the underground alignments; the Kenny Hill formation is a sequence of clastic sedimentary rocks consisting of interbedded shales, mudstones, siltstones, and sandstones. The formation is also known to have undergone mild and regional metamorphic event resulting in changes of sandstone and siltstone to quartzite and schist/phyllite, respectively [14]. It usually weathered to a depth of about 10 m below which the formation is very hard [15]. However, the complexity of the sedimentary residual soils where the parent rock has been subjected to folding has posed considerable difficulties in design and underground construction due to significant changes in ground properties over short distances. Toh et al. [16] and Wong and Singh [17] discussed some engineering properties of the Kenny Hill formation in Kuala Lumpur.

Tan et al. [13] reported some of the challenges encountered in the design and construction of the diaphragm wall in Kenny Hill formation. However, the performance of earth pressure balance (EPB) machine tunnelling in Kenny Hill formation was discussed by Chin et al. [18].

3 Some geotechnical challenges and lessons

The KVMRT Line 1 had successfully overcome significant design and construction challenges in urban underground works, including tunnelling and other construction close to and undercrossing existing structures and buildings, and dealing with issues that had arisen as a consequence of loss of ground and groundwater drawdown. At the same time, a few serious construction problems arisen and while these had largely been overcome, the lessons learned from these incidences are of great help for future construction. Stations and tunnelling works for Line 2 have recently been completed; indeed, because of having more challenging ground conditions and more intricate undercrossing of existing structures, Line 2 will present even greater technical challenges in similar ground conditions.

The following top five geotechnical challenges are identified among these problems for urban underground construction. However, it should be pointed out that the geological challenges/risks are not only the focus of this article. Nevertheless, this article also discussed the lessons learned and experiences gained from the application of innovative techniques and technological advancement in the mitigation of future problems.

3.1 Site investigation and complex ground conditions

There are challenges in interpreting the ground data but even more difficulties in gathering adequate and yet reliable ground data in urban built-up environment. Most site investigations (SI) have constraints in achieving the ultimate objectives with the final scope of works being a compromise between satisfying the constraints and going forward with an acceptable level of ground risks to the underground construction works. The quality and sufficiency of subsurface information obtained for design determines the level of the design risks. Hence, designers would like to have a realistic practical subsurface model with associated design parameters and real-time monitoring.

For a linear infrastructure project like MRT, the SI process starts with boreholes being sunk at about 140 m at the concept design to boreholes as close as 30 to 60 m interval by the time the construction tenders are called and at detailed design stage. During construction, more boreholes are normally sunk to gather even more information. In addition, pre-drilled boreholes carried out prior to the installation of diaphragm wall and SBP wall in limestone formation to delineate and verify ground conditions at karst/slump zones will continue to bring new information about the ground which the designer can make use of.

Figure 8 shows an example of anticipated ground conditions inferred from various stages of ground investigation carried out in one of the Line 2 construction sites. During

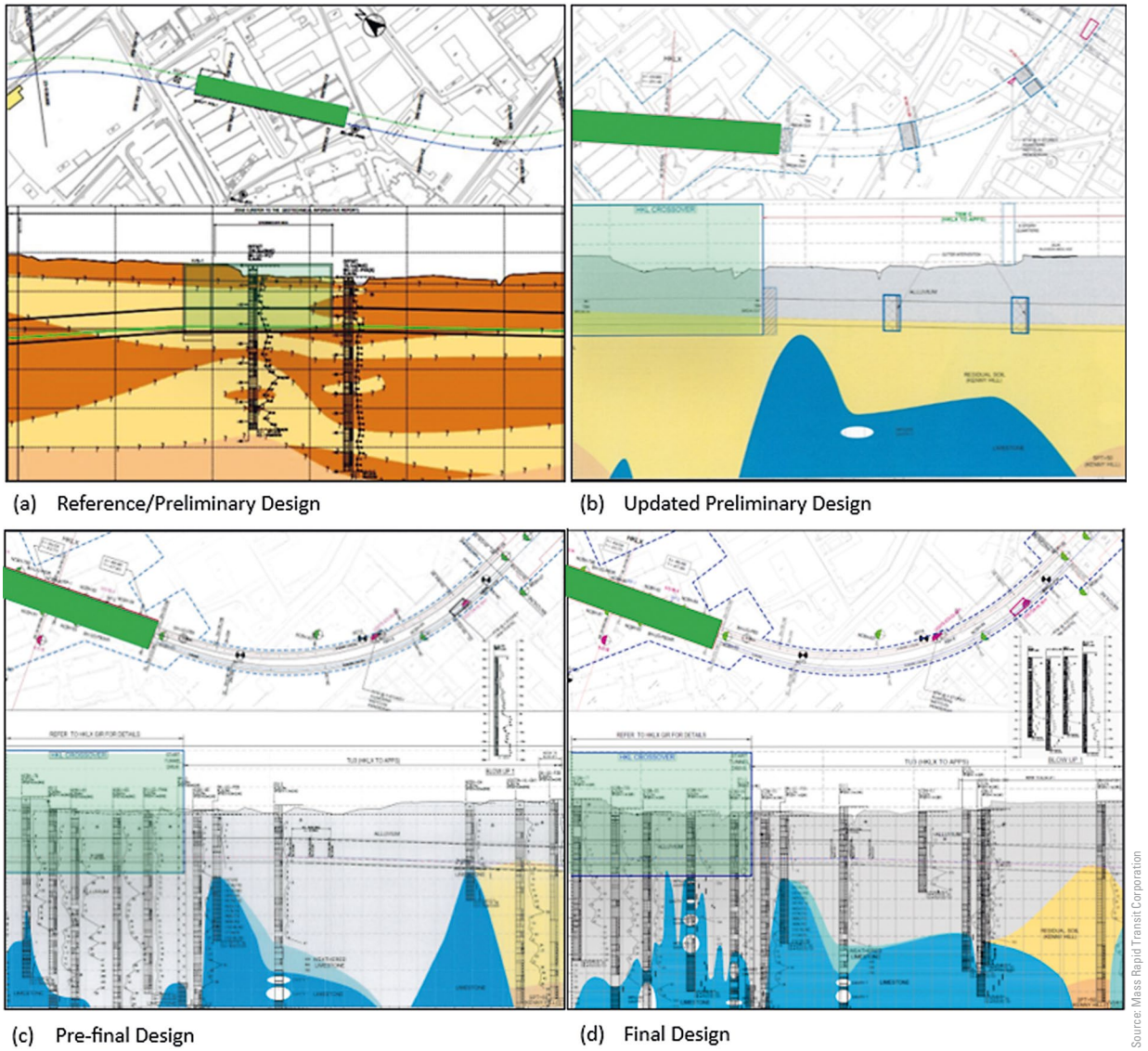


Fig. 8 Interpreted subsurface profiles for various stages of design development: a) reference/preliminary design stage; b) updated preliminary design state; c) pre-final design stage; d) final design stage

the reference design stage, two boreholes were sunk with maximum depth of 50 m. Based on the borehole data, the excavation would be carried out in layered soils, either predominantly clayey or sandy materials, of which overlying the suspected Kenny Hill formation. During detailed design stage, additional SI was carried out with borehole spacing of not more than 50 m along the perimeter of earth retaining system-diaphragm wall. With the new borehole information, it is apparent that the excavation would be in the very thick alluvium soil overlying deep limestone with some locations where the limestone pinnacle raises above the formation level resulting in fair portions of the diaphragm wall requiring socketing into sound limestone bedrock to achieve the adequate design capacity. To a surprise, the ground conditions are very complex, i.e., where the diaphragm wall can be one side rock socketing and the opposite wall terminated in soil (see Figure 9), a condition that may lead to instability and large settlement or sinkhole. Figure 8a to d illustrates the interpreted subsurface profiles for various stages of design development.

Realizing the complexity of the Kuala Lumpur geology, the importance of adequate and reliable ground investigation must not be compromised for underground construction. One of the main challenges is the accurate prediction of the soil-rock interface which is crucial for tunnel operations, i.e., mixed face condition, where the tunnel face is partly in the soil and partly in the rock, a condition that can lead to large settlement or sinkhole formation. One way of gathering more data to better define the soil/rock face during the design stage is to have more boreholes. However, there is a practical limit to this as some of the boreholes would have to be located in places which are not accessible before the construction starts. An alternative is to complement the boreholes with geophysical surveys. From an operational perspective, the purpose of the geophysical survey is to provide a reconnaissance ground investigation with a view to identify areas of enhanced ground risk that may be present and hence to plan an efficient and objective borehole SI programme. However, one must remember that geophysical investigations cannot be a substitute for boreholes but applied cor-

18657389, 2023, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

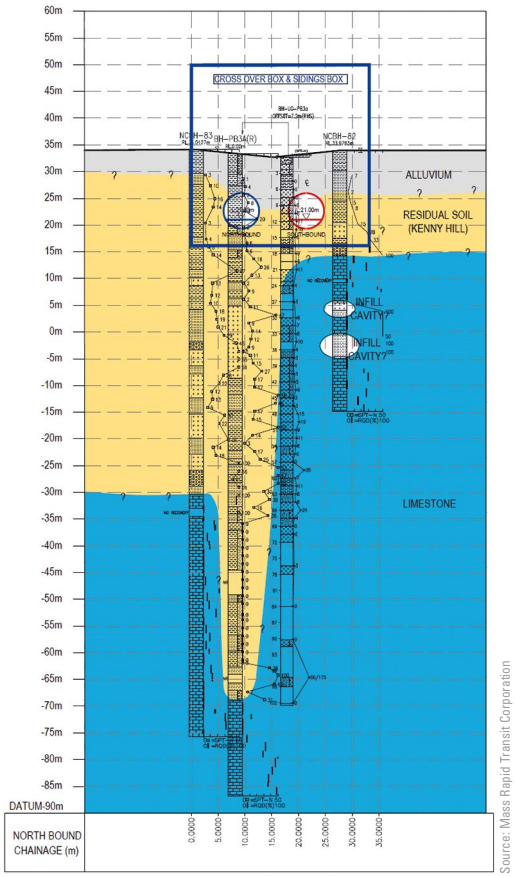


Fig. 9 Geological cross sections indicating the diaphragm wall potentially socketed into rock at one side and terminated in soil at the opposite side

rectly and by observing the many ground constraints of an urban environment, useful support information can be obtained.

Carrying out multichannel analysis of surface wave (MASW) and electrical resistivity (EI) along the proposed underground alignment in the Kuala Lumpur urban areas is a relatively complicated process as a long stretch of made road surface often needs to be temporarily closed and therefore official road closure permits are required. And some areas of geotechnical concern are beneath the existing structures which would present further obstacle, effectively rendering MASW and EI practically impossible. Both methods are significantly susceptible to induced noise, seismic, and electrical interference, which are pervasive in densely built-up and active urban areas. Technically, the results from MASW and EI are subject to influence from the near-surface ground conditions and utilities, and the geotechnical conclusions may be adversely affected accordingly [19].

Nevertheless, gravity survey method was adopted for MRT Line 2. Operationally, the gravity method is a non-intrusive passive technique requiring only a small footprint for setting up, fast in data collection with minimum logistics preparation, and usually demands no special provisions for execution in the urban environment. It may also be conducted within buildings at ground or basement level if necessary and not being confined to profile operations enables lateral definition. Im et al. [19] presented some lessons learnt of the past in the region, and what were the specific measures that had been strictly implemented on this occasion to ensure quality results can be derived from the gravity survey within the urbanized city environment and meet the objectives of the geophysical survey implemented during reference design stage of Line 2. The gravity survey output interpretation model for Chan Sow Lin Station, Figure 10, provides a good example of the application. The geophysical model suggests deep karstic solution conditions, possibly with linear connecting solution channels or large cavities, which would not have been apparent from the two pre-existing boreholes. Figure 11a,b illustrates the geological model before and after the gravity survey and the subsequent controlled borehole investigation. On hindsight, without proper geophysical survey as a guiding tool for repositioning the borehole location, karstic features within the limestone may be frequently unrecognized by adding boreholes in random or conveniently available locations.

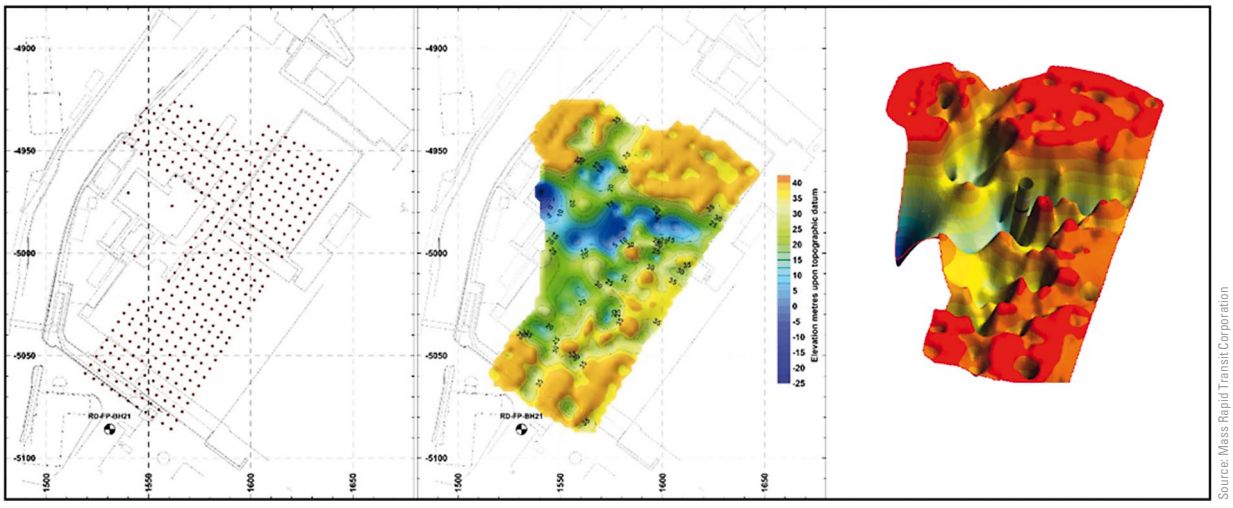


Fig. 10 Gravity survey output interpretation model for Line 2 Chan Sow Lin Station

18637849, 2023, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

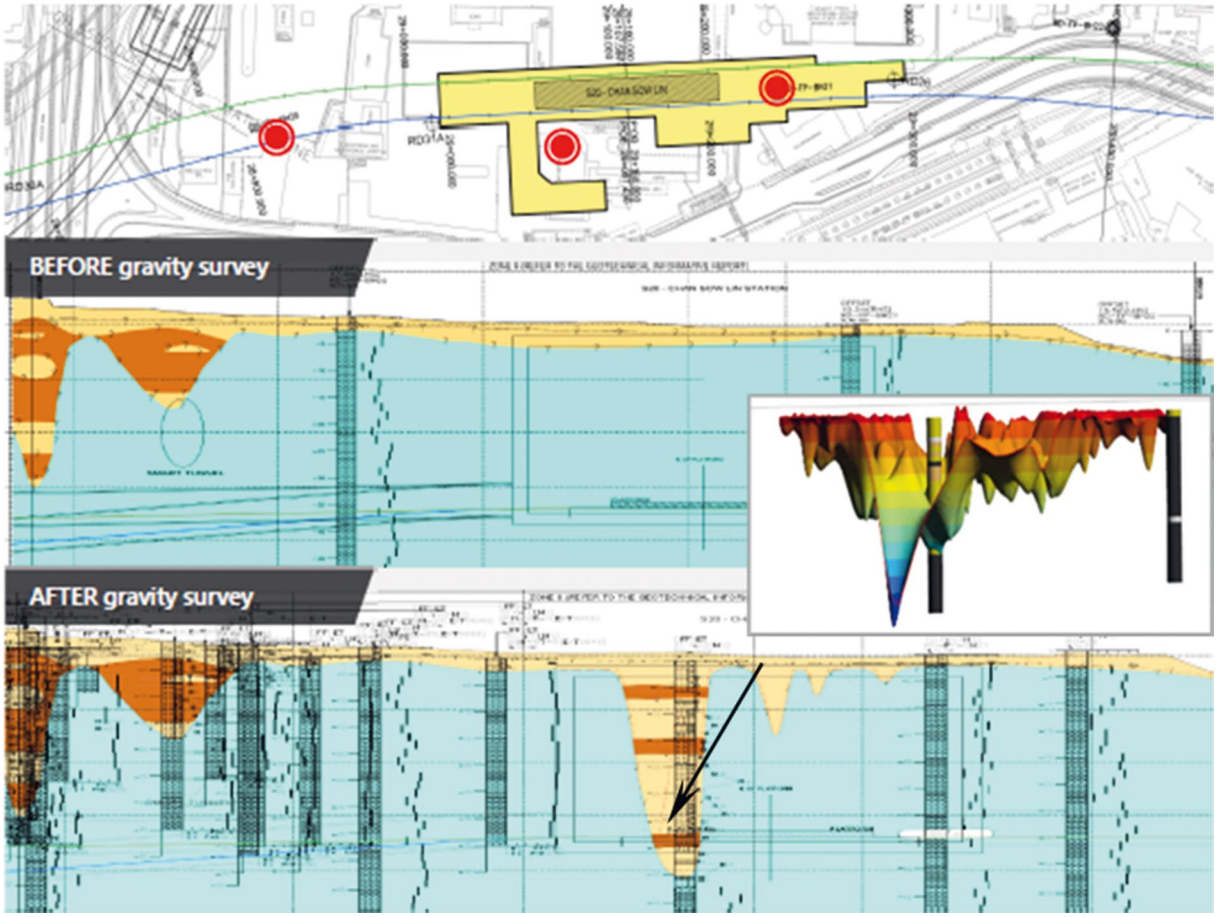


Fig. 11 Case study of gravity survey in Line 2 Chan Sow Lin Station: before gravity survey (top) and after gravity survey (bottom) and verified by borehole investigation at the targeted location

3.2 Design, computational, and analytical development for deep excavation and tunnelling works

Due to the complex behaviour of the subsurface and greater challenges of underground construction in a densely populated urban environment, some prescriptive design criteria are established to reduce risks to acceptably low level and ensure good engineering practices. KVMRT Design Criteria Section 900 and Section 1100 provide good design guidance for foundation and geotechnical works, and tunnel and related works, respectively.

Any deep excavation works would inevitably require some form of groundwater control during excavation. The depths of excavation, the ground conditions, and the land use around the excavation govern the method of groundwater control and groundwater level preservation. Generally, allowable groundwater drawdown is limited to maximum of 1 m below the baseline historical lowest groundwater table and recharging is to be considered with respect to the damage risk to the surrounding buildings whenever this limit is exceeded. Grouting in limestone was carried out as risk mitigation measure for groundwater control to reduce the rate of groundwater inflow into excavation and reduce the pathways of water flow into excavation area. Rock fissure grouting was carried out along the perimeter of excavation area to form curtain grouting up to 10 m below final excavation level. Ha [20] presented case histories of curtain grouting works carried out at Line 1 Cochrane Station and TRX Station (see Figure 12), including the lessons learned and recommendations for further improvement. One strut failure analysis is introduced to ensure the design of earth retaining and stabilizing structure (ERSS) structurally safe, robust, and has sufficient redundancy to avoid catastrophic collapse resulting from an isolated case of failure of any particular member [21].

The designer should carry out risk assessment with the aim to avoid the possibility of such an accidental occurrence through design changes, construction procedural controls, etc. If this does not adequately mitigate the risk, the designer should carry out a calculation to explicitly show that progressive failure will not occur under such circumstances. Greater attention was also given to the design for temporary steel strutting elements. Apart from the earth pressure and groundwater, material dead load, other design criteria such as 1.5 kN/m live load, eccentric load, temperature effect (changes of 10°), accidental impact load (50 kN in vertical direction; 10 kN in horizontal direction), and one-strut failure are considered [22]. Loh et al. [23] discussed the design and construction considerations for steel struts in excavation for KVMRT project.

18657389, 2023, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

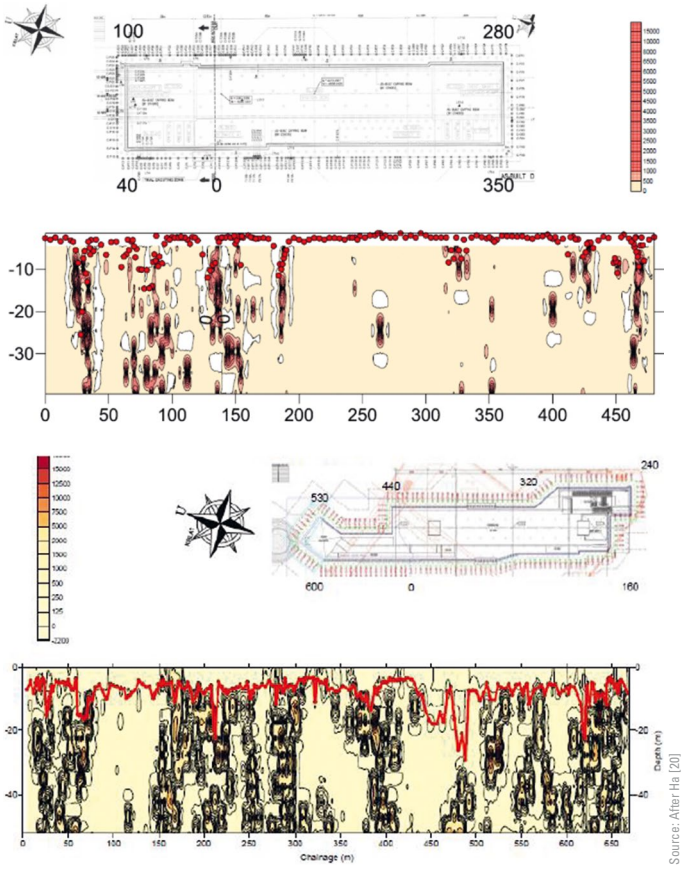


Fig. 12 Elevation of grout take with depth

Needless to say, for today's underground infrastructure demands, geotechnical considerations are coming to the fore with emphasis on solving complex problems by performing numerical modelling and analysis of geo-structures and groundwater flow. It is a clear trend that some of the problems that previously were limited to 2D plan strain or axisymmetric idealizations now are analysed by full 3D models. The use of 3D analyses will in several cases give more representative results and thus more realistic and safer solutions when compared to using the traditional methods and also compared to 2D finite element (FE) analysis. Having said that, the most challenging part of geotechnical FE analyses is to properly describe the constitutive behaviour of the soil. Different models may be used for the calculation of displacements, load-carrying capacity and soil-structure interaction, undrained/drained behaviour, etc. Hence, appropriate design of underground structures demands proper understanding of the behaviour as well as sufficiently good models that may simulate the rightful behaviour of natural ground. Tan et al. [3] summarized that HS-Small model, which incorporates small strain stiffness, offers potential savings in the design of deep excavation works and represents a step forward in the understanding of soil-structure interaction. This has been re-affirmed by Cheah et al. [24] in the case study of Line 1 Bukit Bintang Station, as shown in Figure 13. They further demonstrated that building stiffness needs to be taken into consideration in the numerical modelling on top of the small stiffness of soil in order to obtain a reasonable agreement with the measured building settlement.

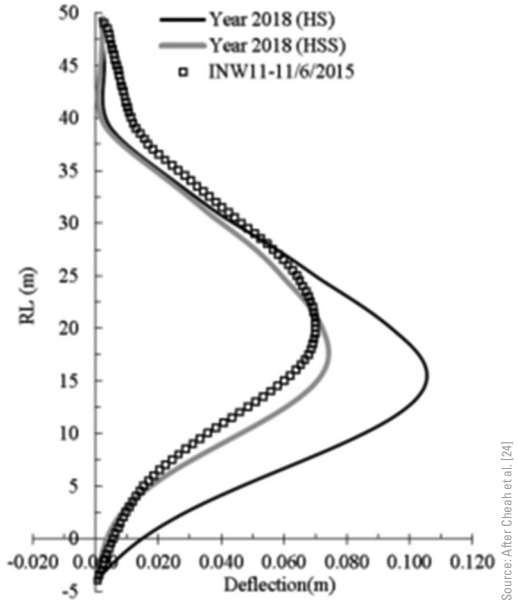


Fig. 13 Comparison of measured and predicted wall deformations

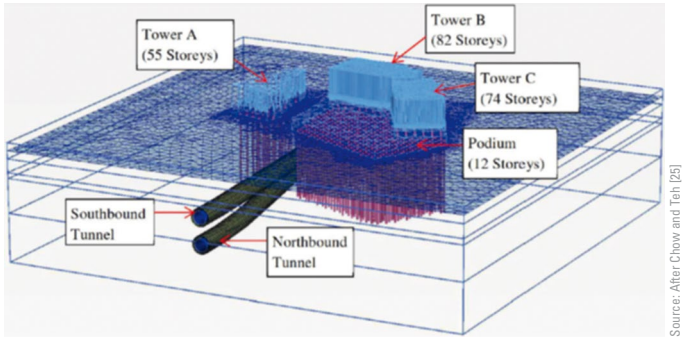


Fig. 14 3-D FE modelling

More recently, 3D FEM is widely used to model the entire tunnel construction sequence for a more rigorous analysis and assessment of the potential impact due to top-side development. For practical applications, the top-side buildings usually straddle over the tunnels of which the development works shall comply with the requirements of Railways (Railway Protection Zone) Regulations 1998. Chow and Teh [25] carried out extensive 3D FE analyses to ensure compliance with the statutory requirements, where the Line 2 tunnels run across the proposed high-rise development in limestone formation (Figure 14).

3.3 Reliable predictions of ground movements and damage risk assessment

Deep excavation and tunnelling work generally induced ground movements in different magnitude. Therefore, in the highly populated urban environment of Kuala Lumpur, a systematic process of ground movement prediction, building impact/damage risk assessment, and protection is required. Comprehensive procedure of ground movement prediction and building damage risk assessment applied in KVMRT project was broadly discussed by Boon et al. [26].

18637849, 2023, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

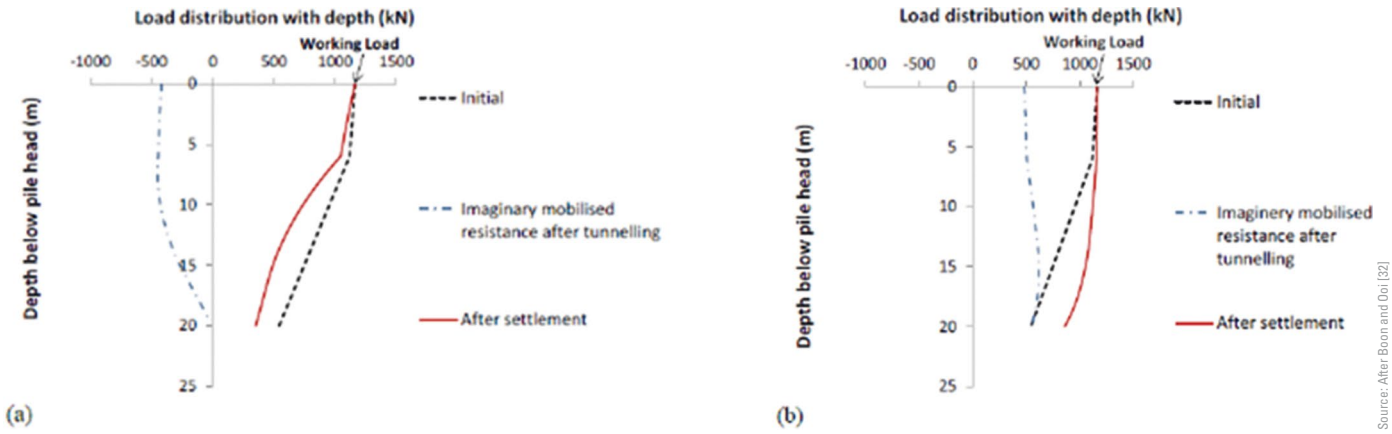


Fig. 15 Load redistribution with depth due to tunnelling-induced settlement

To examine the influences on adjacent ground, utilities, and structures due to the underground construction, ground movement analysis was performed first. The evaluation was conducted using empirical methods or numerical modelling. Its potential impact is then assessed using the three-stage risk assessment approach recommended by Mair et al. [27]. Different degrees of geotechnical input and analyses are applied depending on the level of risk and the stage of the project. For Line 2, a risk-based (index system) approach similar to that suggested by Goh and Mair [28] is adopted to consider the pre-existing structural/building conditions on top of damage categories suggested by Burland et al. [29] and Boscardin and Cording [30].

For buildings assessed to be at a high risk of damage, detailed evaluation is to be undertaken. This could involve evaluating the structural details of the building, giving full consideration of the construction method in three dimensions rather than plane strain, as well as including soil-structure interaction effects which means taking into account of the building stiffness. There is a propensity for buildings to re-distribute the tunnelling-induced movements such that stiffer buildings would experience much less differential settlement than flexible buildings—this has been illustrated by several researchers and more recently by Mair [31]. Notwithstanding, there is considerable anxiety to such tunnelling activities either by a laymen or local engineers, and understandably so because urban tunnelling is still relatively new to the local fraternity. There were concerns over the loss of support directly below the building foundations, thereby causing severe building damage. This is also aggravated by the dearth of local published information related to tunnelling directly below buildings.

Through analytical studies, Boon and Ooi [32] showed the use of load transfer method to evaluate the redistribution of load in piles with depth caused by tunnelling-induced settlement with greater certainty, as shown in Figure 15. Mair and Williamson [33] reported from centrifuge observations that there are only relatively small changes in load distribution in the pile during tunnelling, with some reduction in pile skin friction under negative relative displacements and an increase in pile skin friction under positive relative displacement. Pile failure does not occur even at high tunnel volume loss and there is little or no loss of capacity of the piles. Concerns about loss in pile-carrying capacity are often misplaced and with appropriate tunnelling controls, it is possible to keep the building movements within the normal range of ground settlements expected.

For the first time in the construction of stacked twin tunnels, with the upper tunnel constructed first and subsequently undermined by the lower tunnel in Pudu shaft of Line 1, Boon et al. [34] presented some insight into the measurements and analytical assessment of this case history. They showed that the longitudinal settlement profile of the upper tunnel could be analysed as a Winkler beam. The Winkler beam analysis with the assumption of loss of support pressure behind the TBM shield resulted in good agreement with field measurements with regard to the deflected shape of the upper tunnel at different points of interest with respect to the lower TBM face (see Figure 16). The extensive data collected from the instrumentation during the construction of Line 1 tunnels have been recently used by Khoo et al. [35] to study the actual volume loss caused by tunnelling in Kenny Hill Formation. Further attempts to establish links between the key parameters of the tunnelling process, such as face pressure as well as tail void grouting pressures and the measured volume losses, were presented in their subsequent article [36]. The focus has been to observe general trends,

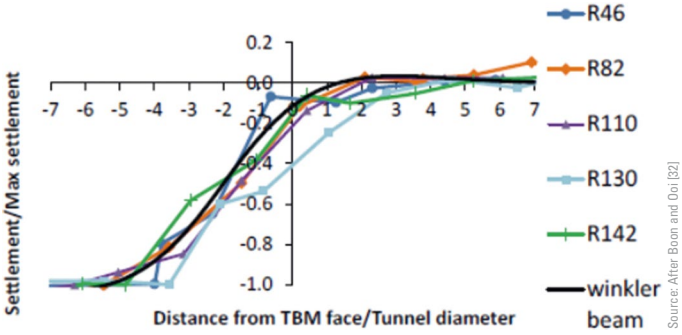


Fig. 16 Longitudinal settlement profile obtained by using a specific ring number as a reference point and measuring the distance of the TBM face

18657389, 2023, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

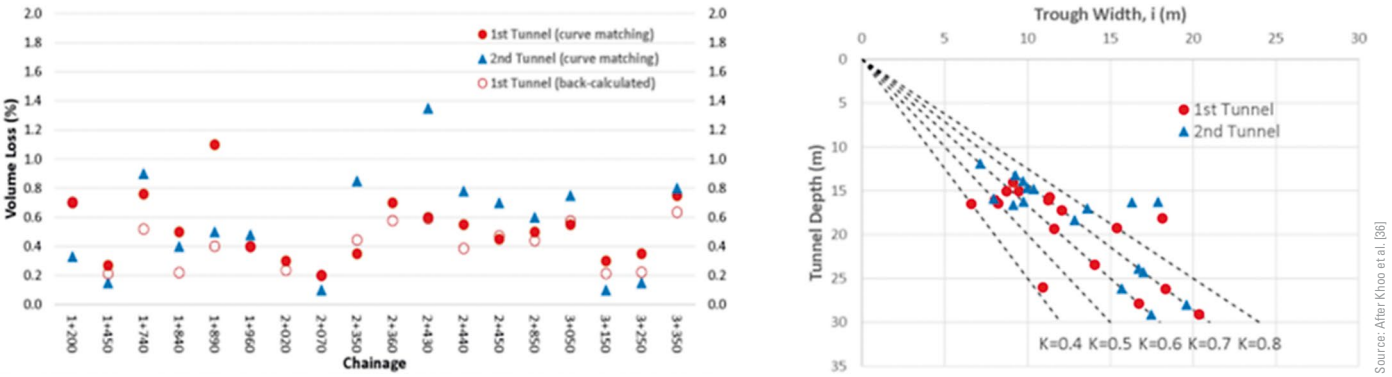


Fig. 17 Back-analysed volume loss and trough width parameter for EPB tunnelling in Kenny Hill Formation

with the objective of assessing the effectiveness of face pressure and tail void grouting in reducing volume loss.

The results in Figure 17 show that tunnelling in Kenny Hill formation has achieved a volume loss ranging from 0.10 % to 1.35 %. However, the majority of the volume loss is below 1.0 %. Trough width parameter, K, between 0.5 and 0.7 was obtained with an average value of 0.6. Khoo et al. [36] also reported that the majority of the construction settlement is associated with the tail void (resulted in ⅓ to ⅔ of the total volume loss) and that surface settlement directly above the funnel face is generally much less than $0.5S_{max}$. The results provide valuable knowledge on ground responses to tunnelling in Kenny Hill formation.

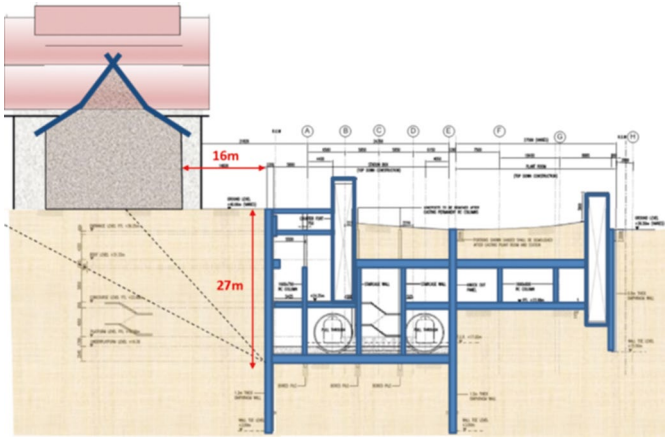


Fig. 18 Critical cross sections showing deep excavation adjacent to Museum building

3.4 Development of protective measures for the existing urban setting

For public convenience, the underground stations are generally constructed increasingly close to or even directly beneath existing buildings/roads in congested city. The construction of station near an existing building may induce significant building settlement, not only in the transverse but also in the longitudinal direction.

Following the detailed evaluation (impact assessment), consideration is then given to protective measures needed for buildings that remain in the high damage risk categories. Mahalingam et al. [37] presented the design management and case study of underground construction for Line 1 Muzium Negara Station in close proximity to National Museum building (see Figure 18). From the onset of the project, the potential problems faced by the building were identified and a carefully risk-managed approach for settlement control and minimizing construction impact was undertaken together with extensive instrumentation monitoring of the structure and the surrounding ground. The station construction scheme initially planned using the bottom-up method with multiple levels of temporary strutting was, however, changed to a top-down method using diaphragm wall as both temporary and permanent earth retaining system. Khoo et al. [38] provided a detailed account of the actual performance of the settlement sensitive heritage Museum building in response to the deep excavation in close proximity.

One of the biggest challenges in undertaking underground construction in a highly urbanized environment is to tunnel directly below buildings and their foundations. Boon et al. [26] gave the general framework that was adopted to identify and predict performance of the buildings and structures within the influence of tunnelling. Once the risk has been fully evaluated, the actual response of the structure to the tunnelling works is carefully monitored and evaluated against the design prediction.

Clearly, Line 1 had successfully tunnelled directly below a few buildings and infrastructures using TBM. Boon's article discusses the Sungai Kerayong bridge case history, in which the tunnel drives encroach into the foundation systems of a road bridge and the micropile retaining walls along a river. Meanwhile, tunnelling below the existing old road bridge was reported by Mak et al. [39]. In the absence of foundation as-built records, field investigation and verification were commenced with trial pit to excavate below the pile cap to determine the type of pile. Boreholes were then carried out next to the pile to determine the ground condition supporting the piles followed by parallel seismic survey to determine the pile length.

18637849, 2023, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

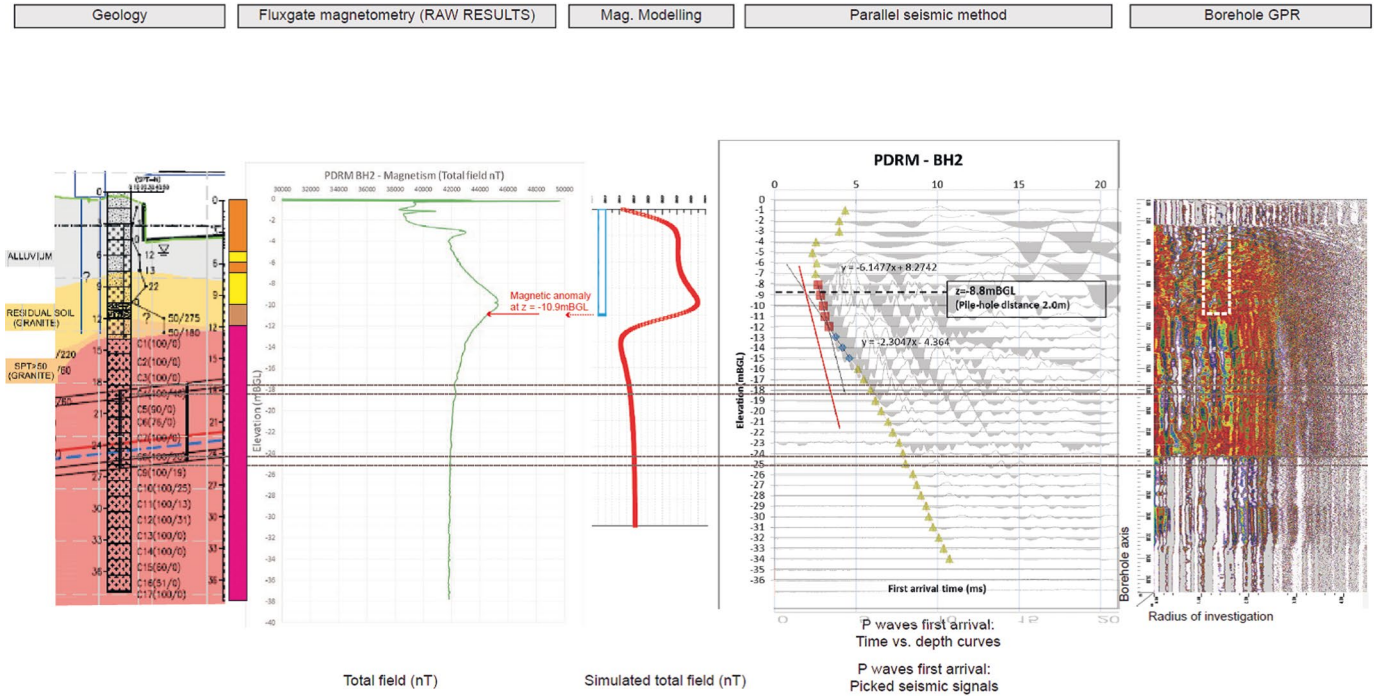


Fig. 19 Comparison of pile length determination from magnetometry, parallel seismic, and borehole radar methods

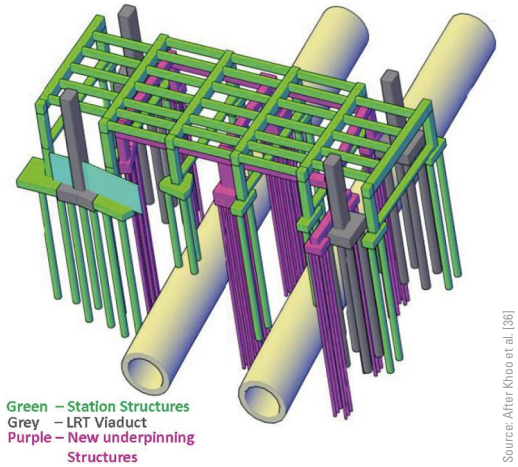


Fig. 20 3D perspective of the underpinning structures with undercrossing bored tunnels

The same borehole may be used to detect the likely depth of the pile using other geophysical methods such as magnetometry or borehole radar measurements (see Figure 19 for a sample comparison). It has to be noted that both parallel seismic method and borehole radar method are not effective in identifying the socketed section of the pile due to lack of contrast in stiffness between the pile and bedrock [20]. The inferred pile length was then used for clash analysis and also to evaluate the impact of tunnelling on the pile performance.

Khoo et al. [40] reported the challenges of tunnelling below the operating Pasar Seni LRT Station where extensive enabling works, such as structural underpinning, construction of temporary shafts, and mined adits for pile removal, were carried out prior to TBM tunnelling. Figure 20 shows the 3D perspective of the underpinning structures with undercrossing bored tunnels.

3.5 Implementation of complex monitoring system

It is almost impossible to fully define the subsurface condition and its response to underground works for a design hypothesis. As a result, instrumentation and monitoring forms the vital link between the design hypothesis and actual performance of construction works as well as a crucial component of the risk management procedure. As construction progresses, exact geotechnical observations and behaviour monitored using instrumentation assisted construction team and designers to make a judgement, evaluate, and make changes to construction methodology or design parameters, if necessary. It plays an important role in providing timely information to the entire construction and design team, client, stakeholders, and public at large in order to ensure safe and smooth operation of the underground construction besides legal protection and related economic issues.

Various types of instruments were placed on the surface and subsurface to monitor the surrounding ground responses and structures within the influence zone during tunnelling and excavation works. According to Vasagavijayan et al. [41], close to 20,000 instruments were installed and monitored during Line 1 underground construction. All monitored geotechnical parameters were observed and compared with pre-established design threshold values defined as AAA levels (Alert, Action, and Alarm). If there is any reading that breached the AAA level, prompt notification will be sent to relevant parties immediately so that required mitigation measure can be carried out in a timely manner. A well-documented control procedure was put in place to ensure all involved in the project are receptive to the instrumentation monitoring results and work proactively to ensure safe job site and surroundings.

18657389, 2023, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

1865864, 2023, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

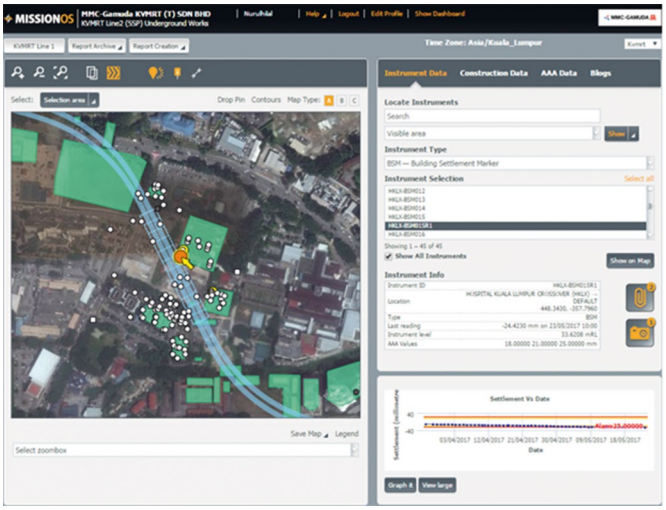


Fig. 21 Web-based online IDMS

Due to the mega scale of this project which involves thousands of data to be assimilated daily, a data management system capable of displaying information with the added functionality such as alert capability is crucial in creating an effective monitoring system so that proper action/counteractions can be implemented in a timely manner. With many millions of records and gigabytes of data, this can only realistically be achieved using a web-based online monitoring system which is also capable of managing large capacity data from real-time monitoring instruments as well as TBM driving parameters and progress. Maxwell et al. [42] demonstrated the contribution of the centralized instrumentation data management system (IDMS) to assist the underground construction in KVMRT project. This system allows multiple users to login simultaneously and can be accessed via computer or even smart phone provided they are connected to internet. A caption of this online monitoring screen is shown in Figure 21.

The IDMS brings together information from ground investigation, design, production, and monitoring. Figure 22 shows detailed view of progressing tunnel undercrossing existing building structures and contoured layers of best estimate settlement. With the data, spatial analyses can be undertaken in terms of proximity to construction faces; they are useful tools for analysing settlement development around the tunnels in varying ground conditions. Large quantities of data assist engineers to study both the effects of tunnelling and the potential causes.

The IDMS has allowed all parties, client, consultants, and contractors to access real-time monitoring data as well as the construction activities carried out at site through a common platform that provides interactive functions to generate graphical plots linking the monitoring data and the construction activities [43].

4 Some other innovations in urban underground construction

This section provides a brief overview of some new innovative ideas, creative thoughts, and the state-of-the-art technologies that support the underground construction in KVMRT project. Highlighted are geotechnical solutions and geotechnologies that provide opportunities for significant improvement in the delivery of the cost-effective design and contribute to resilient and sustainable urban underground construction. Such innovation takes place through the provision of more effective designs, processes, technologies or construction methodologies that 'breaks into' the industry. Indeed, innovation often manifests itself via the engineering process employed when the problem being solved is of a technical or scientific nature.

4.1 Innovative design and construction methods

Khoo et al. [44] presented an innovative design and construction approach for mined adit to facilitate pile removal work at existing operating Pasar Seni LRT Station.



Fig. 22 Progressing tunnel-ring colour shows weekly progress (left). Contoured layers of best estimate settlement (right)

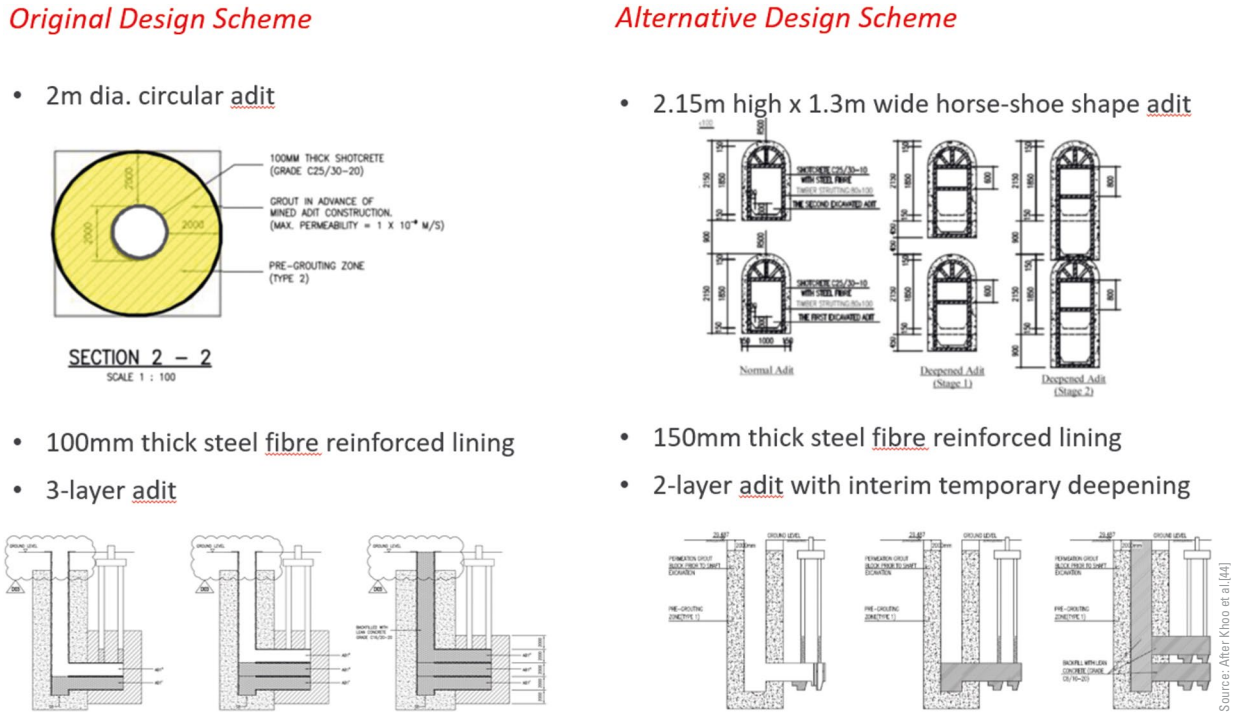


Fig. 23 Horizontal mined adit for pile removal

Three layers of conventional circular adit were originally proposed to provide connecting access from the caisson shaft for the pile cutting. With the recent experience in deep excavation and tunnelling in Kenny Hill formation and comprehensive engineering analyses, the adit was alternatively proposed in horseshoe shape and changed to two-layer with interim deepened excavation supported by timber frame. The alternative adit scheme has successfully implemented with substantial cost and time saving from the original design (Figure 23).

In Line 2, the pile removal method is further improvised in order to ensure all obstacle piles are removed prior to tunnel arrival within a tight schedule. The affected pile sections will be removed by inclined coring from the ground surface. Tan et al. [45] presented the case study to remove partially the existing 300 mm RC square piles supporting the education quarter building at Kampung Bharu to allow for tunnel boring. The coring direction and distance has to be planned thoroughly considering the existing site constraints. In order to ensure stability of drilled hole during pile coring, ground improvement by means of jet grouting block was carried out surrounding the edge of the inclined drilled hole (see Figure 24). However, it should be noted that this pile removal method may not be as complete or thorough as compared to mined adit method. This is considered manageable for the tunnel operation as long as the majority of the pile material, especially the pile reinforcement, can be removed preventing jamming of TBM system.

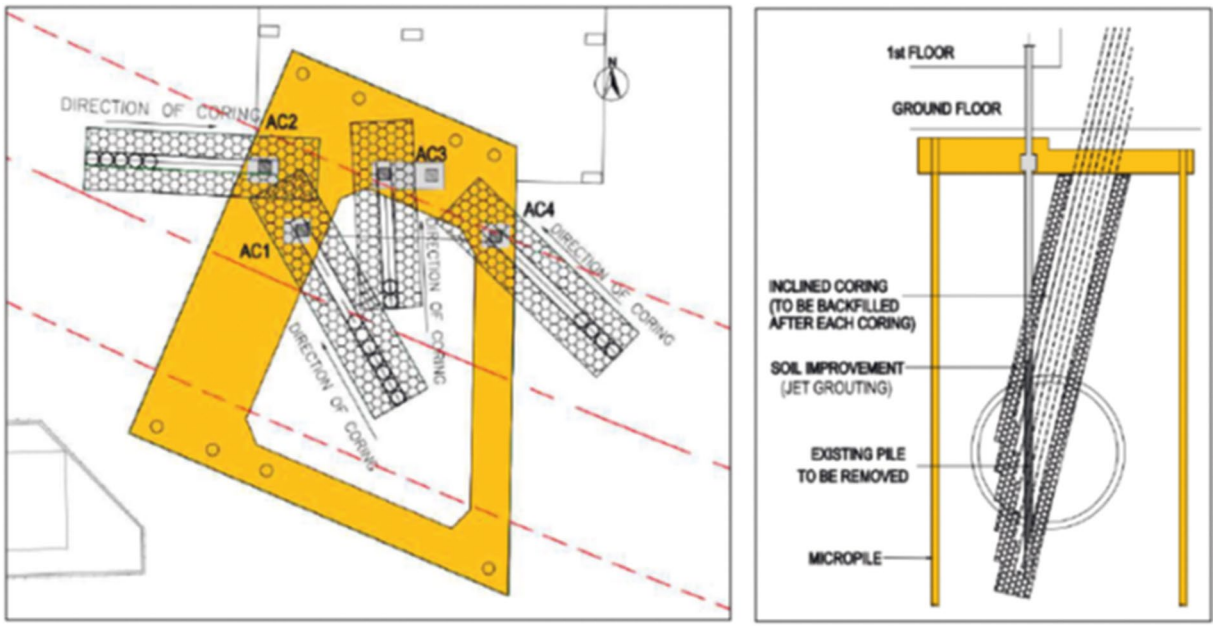


Fig. 24 Pile removals from ground surface using coring rig layout plan (left) and typical section (right), after Tan et al. [45]

18657389, 2023, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

18637849, 2023, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

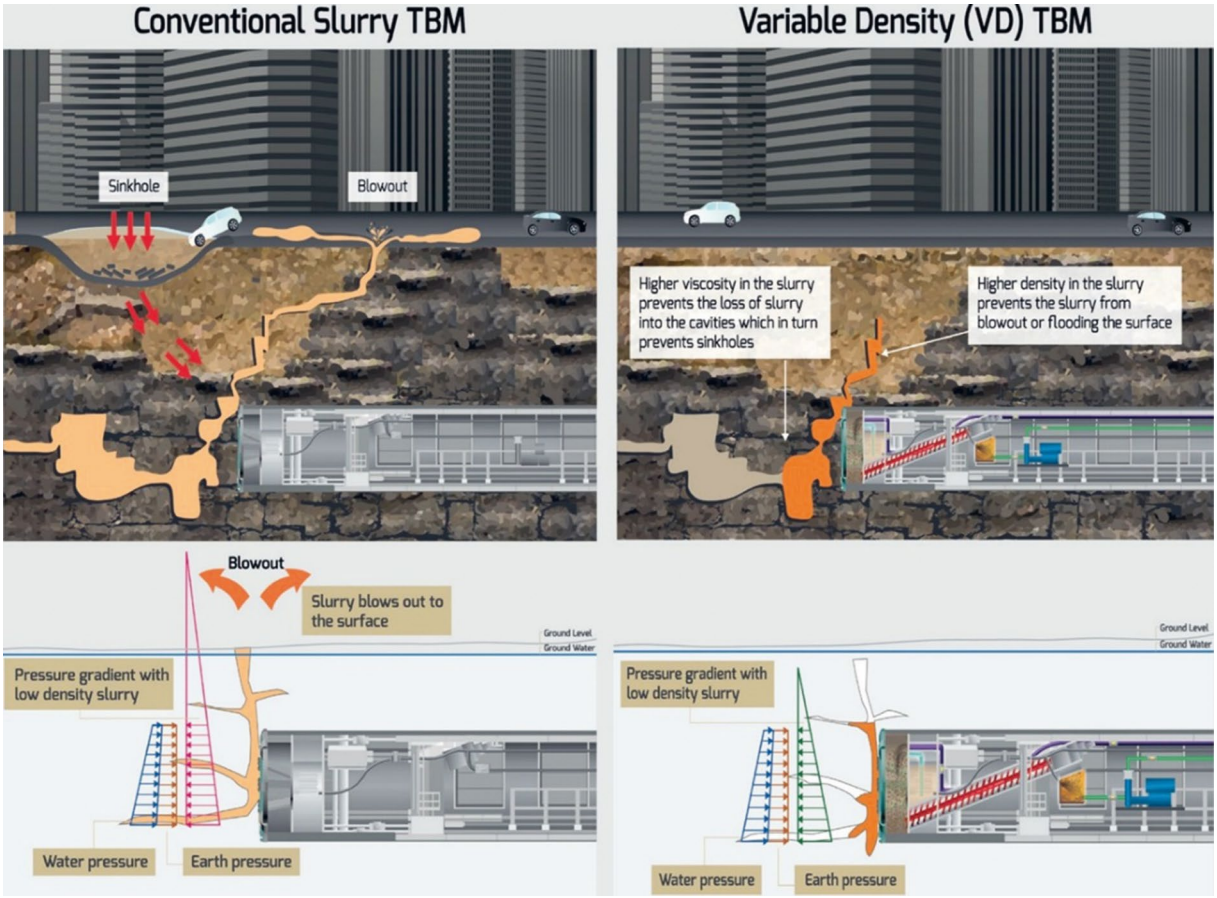


Fig. 25 Conventional slurry TBM versus variable density TBM

4.2 Innovative in underground technology

Needless to say, the variable density tunnel boring machine (VD-TBM) born out from the ingenuity of the local tunnelling contractor, MMC-Gamuda in collaboration with Herrenknecht, specifically designed for coping with the specific challenges presented by Kuala Lumpur's extreme karstic limestone geology, has demonstrated the fact that Malaysian professionals have not only learned from the training, but, most importantly, have developed their expertise to come out with ingenious and innovative solutions to their underground problems. This technological innovation is the first of its kind in the world and it played as an important milestone of solving the tunnelling problems in Kuala Lumpur karst and has far reaching effect on the development of tunnelling engineering in Malaysia [46] (see Figure 25). The conventional slurry TBM may frequently encounter cavities or fissures in karstic conditions into which the slurry would escape, resulting in high volume and pressure loss. The variable density slurry shield enables the TBM to alter the density and viscosity of the slurry according to soil condition when tunnelling. Slurry of higher viscosity can stop slurry of lower viscosity from escaping into cavities or blowing out from fissures leading to the surface. This, in turn, preserves the face pressure of the TBM, keeps the terrain stable during the excavation process, and prevents upward flows of slurry, reducing the risk of slurry fountains on the built-up surface [47].

The VD-TBM can be operated in four different tunnelling modes and thus perfectly adapted to different geological conditions. It is a further development of the multi-mode TBM, which combines the advantages of the EPB and slurry-supported (Mixshield) modes. In all modes, the excavated material is removed from the pressurized chamber by a screw conveyor. In EPB mode, the screw conveyor speed and the advance rate control the face support pressure. The screw conveyor drops the excavated material onto a belt conveyor. In slurry mode, the face support pressure is automatically controlled by an air cushion. The pressure communication between working and excavation chamber takes place by means of two pressure lines, or communicating pipes. In addition, in slurry mode it is possible to vary the density of the support medium in the excavation chamber. The choice between the more fluid low-density support medium (LDSM) and the denser high-density support medium (HDSM) considerably increases the areas where variable density can be used. The high-density mode closes the gap between Mixshield and EPB operation. Further details can be referred to ref. [47].

The variable density TBM has proven to be an effective excavation tool in complex geologies and performed beyond expectations. It has since received international recognition from tunnelling fraternity and accorded the Technical Innovation of the Year Award in 2014 by NCE and ITA. As an extent of the VD-TBM, the team devel-

oped the autonomous TBM (ATBM) system, which is a 'plug and play' solution imagined as a custom software installed on an industrial PC module, has won the Technical Production/Equipment Innovation of the Year award at the 2019 ITA Tunnelling Awards. The ATBM stands out as a novel approach to tunnelling as the world's first automated system comprising custom artificial intelligence control algorithms enabling autonomous control of TBMs.

Geotechnical instrumentation provides the basis for understanding ground response to underground construction and the impacts on existing structures and the natural environment. As a general trend, advances in instrumentation monitoring are moving away from physical measurements at a limited number of points, to widely distributed, wirelessly connected sensor networks, and to areal scanning techniques. These data allow engineers to optimize design, construction processes and increase safety on the job. In KVMRT project, new instruments and technology such as fibre optic sensing system have been widely used in the instrumented preliminary pile load test to provide insightful results [48, 49]. More recently, fibre optic sensing is placed in cast in situ pile for continuous in-pile instrumentation which allows the bending moment and axial force developed during tunnelling to be monitored in real-time manner. The research will add to the existing knowledge of field monitoring of pile responses during tunnelling. Hisham et al. [50] also presented the potential of using fibre optic sensor in monitoring the load development and displacement of tunnel lining.

Opportunities to advance construction monitoring include the means to collect and analyse detailed real-time monitoring data during construction equipment operation. For instance, cutterhead operation data (e.g., torque, power, thrust, penetration per revolution, disc cutter forces, rotational speed, and bearing temperature) accessible from the IDMS increase knowledge of excavation processes, in particular geological conditions. In a similar manner, continuous measurement of the volume, weight, and advance rate of removed spoil is an important indicator of ground displacement and settlement control. Integrating data from Line 1 and Line 2 can facilitate improved equipment design, performance prediction, and understanding of zonation and geologic material properties.

5 Conclusion

Given the large-scale underground construction is being implemented in Kuala Lumpur, geotechnical engineering in the area of deep excavation and tunnelling is also being given high attention in recent years. While economical and expeditious construction methods are consistently sought to keep underground project viable, the densely developed urban areas with variable ground conditions pose great challenges for construction in ensuring that the safety and convenience of the public are not compromised and the damage to adjacent structures is mitigated. Some geotechnical challenges in both design and construction of deep excavation and tunnelling works and some of the salient aspects of the underground construction that are unique to KVMRT project have been presented. Geotechnical applications have supported the design and construction of underground stations and tunnels, and will continue to be critical to the delivery of underground works with lower initial costs and risk, and better lifecycle performance.

The knowledge gained and lessons learned from the KVMRT Line 1 and Line 2 are being effectively transferred to the design of Line 3 which is expected to have even greater technical challenges. Constructing Line 3 will become more challenging as the excavations and tunnelling works will likely get deeper and closer to very densely populated areas requiring extensive interchanges with existing stations, crossing multiples existing infrastructures and live rail tunnels as well as integration with existing or future developments.

Improvements in underground technologies have enabled great strides in urban development in recent decades, but the complexity and unpredictability still inherent in underground construction are indications that many challenges remain. There are opportunities for new methods and technologies to address these issues and build a better sustainable underground development. Leveraging new digital workflows which incorporate real-time monitoring and analytics during infrastructure operations, geotechnical professionals can play the increasingly valuable role they deserve in achieving geo-environmental resilience.

References

[1] Suruhanjaya Pengangkutan Awam Darat (2010) *Urban rail development plan*. Malaysia.

[2] Azlan Marzuki Shah, J.; Abbott, G. A. (2015) *Risk management for underground works*. Proceedings of International Conference and Exhibition on Tunnelling and Underground Space (ICETUS 2015), Kuala Lumpur, pp. 192–196.

[3] Tan, Y. C.; Chow, C. M.; Koo, K. S.; Nazir, R. (2015) *Challenges in design and construction of deep excavation for KVMRT in Kuala Lumpur limestone formation*. Jurnal Teknologi (Sciences & Engineering), Penerbit UTM Press Malaysia.

[4] Yee, Y. W.; Tan, Y. C. (2015) *Excavation support for TBM retrieval shaft using deep soil mixing technique, Kuala Lumpur*. Proceedings of International Conference and Exhibition on Tunnelling and Underground Space (ICETUS 2015), Kuala Lumpur, pp. 272–278.

18657389, 2023, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TOPICS Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[5] Tan, Y. C.; Koo, K. S.; Ting, D. I. (2019) *Deep intervention shaft excavation in Kuala Lumpur limestone formation with pre-tunnelling construction method*. Tunnels and Underground Cities: Engineering and Innovation meet Archaeology, Architecture and Art. Taylor & Francis Group, London, ISBN 978-1-138-38865-9, pp. 1579–1588.

[6] Ooi, T. A.; Khoo, C. M. (2017) *Tunnelling activities in Malaysia – A review*. Proceedings of Southeast Asian Conference and Exhibition in Tunnelling and Underground Space (SEACETUS 2017), Kuala Lumpur, pp. 243–252.

[7] Poh, S. T.; Low, Y. Y.; Ooi, L. H. (2014) *Steel fibre reinforced concrete tunnel lining for a metro project – A Malaysian experience*. Proceedings of 12th International Conference on Concrete Engineering & Technology (CONCET 2014), Subang, Selangor.

[8] Lim, H. Y.; Ng, H. W. (2015) *Steel fibre reinforced concrete tunnel lining segments*. Proceedings of International Conference and Exhibition on Tunnelling and Underground Space (ICETUS 2015), Kuala Lumpur, pp. 198–206.

[9] Yin, E. H. (1986) *Geology and mineral resources of Kuala Lumpur-Klang Valley (Draft)*, Geological Survey Malaysia District Memoir.

[10] Klados, G.; Ng, H. W.; Wong, G. C. Y.; Chin, J. J. H. (2015) *Managing tunnelling challenges through Kuala Lumpur karst formation using Variable Density TBMs*. Proceedings of International Conference and Exhibition on Tunnelling and Underground Space (ICETUS 2015), Kuala Lumpur, pp. 186–191.

[11] Khoo, C. M. (2015) *Report on talk "Tunnel boring and other underground works of the KVMRT SBK Line" delivered by Gus Klados*. IEM Bulletin, Feb. 2015, pp. 28–29.

[12] Goh, C. Y.; Lim, Y. S.; Ong, K. C. (2015) *Challenges of a Deep Excavation at an Urban Area*. Proceedings of International Conference and Exhibition on Tunnelling and Underground Space (ICETUS 2015), Kuala Lumpur, pp. 250–255.

[13] Tan, J. G.; Ooi, L. H.; Yeoh, H. K. (2015) *Considerations of deep excavation in Kenny Hill and Kuala Lumpur limestone formation*. Proceedings of International Conference and Exhibition on Tunnelling and Underground Space (ICETUS 2015), Kuala Lumpur, pp. 208–218.

[14] Mohamed, Z.; Rafek, A. G.; Komoo, I. (2007) *Characterisation and classification of the physical deterioration of tropically weathered Kenny Hill rock for civil works*. Journal of Geotechnical Engineering 12, p. 16.

[15] Ooi, T. A. (1986) *Design and construction problems of foundation for highrise structures in the Kuala Lumpur Areas*. Proceedings of IEM-ISSMFE Joint Symposium on Geotechnical Problems, Kuala Lumpur, pp. 112–122.

[16] Toh, C. T. et al. (1989) *Design parameters for bored piles in a weathered sedimentary formation*. Proceedings of 12th International Conference on Soil Mechanics and Foundation Engineering, Rio de Janeiro, Vol. 2, pp. 1073–1078.

[17] Wong, J.; Singh, M. (1996) *Some engineering properties of Kenny Hill formation in Kuala Lumpur*. Proceedings of 12th South-East Asian Geotechnical Conference, Kuala Lumpur, pp. 179–187.

[18] Chin, R. M.; Salehudin, M. S.; Abdullah Husairi, N. (2015) *Performance of EPB tunnelling in Kenny Hill formation*. Proceedings of International Conference and Exhibition on Tunnelling and Underground Space (ICETUS 2015), Kuala Lumpur, pp. 298–301.

[19] Im, C.; Collar, F.; Poh, S. T.; Khoo, C. M. (2016) *Application of gravity survey in Kuala Lumpur MRT Line 2*. Proceedings of Underground Singapore, Singapore, pp. 235–247.

[20] Ha, P. T. T. (2018) *Underground MRT in Kuala Lumpur – The inevitable urban transit solution*. Journal of The Institution of Engineers 79, No. 1, pp. 2–20.

[21] CIRIA 760. *Guidance on embedded retaining wall design*.

[22] Mass Rapid Transit Corporation. (2011) *KVMRT design criteria*.

[23] Loh, Y. E.; Liew, S. S.; Tan, Y. C.; Tong, H. S. (2014) *Design and construction considerations for steel struts in excavation of KVMRT project in Malaysia*. HKIE-IEM-CIE Tripartite Seminar on Recent Developments in Limit State Design for Geotechnical Works, Hong Kong.

[24] Cheah, F. L. P.; Law, K. H.; Mohd Ismail, M. A. (2018) *Finite element analysis of a deep excavation in Kenny Hill formation with consideration*. IEM-CIE-HKIE Tripartite Seminar on Geotechnical Challenges in Infrastructure and Transportation Projects, Putrajaya, Malaysia.

[25] Chow, C. M.; Teh, W. S. (2019) *3-D finite element analyses on interaction between building foundations and tunnels in soil and rock*. Proceedings of 1st Malaysian Geotechnical Society and Geotechnical Society of Singapore Conference, Petaling Jaya, Malaysia.

[26] Boon, C. W.; Teh, E. H.; Ooi, L. H. (2015) *Protection of buildings and structures within the influence of tunnelling works*. Proceedings of International Conference and Exhibition on Tunnelling and Underground Space (ICETUS 2015), Kuala Lumpur, pp. 256–262.

[27] Mair, R. J.; Taylor, R. N.; Burland, J. B. (1996) *Prediction of ground movements and assessment of risk of building damage due to bored tunnelling*. Proceedings of International Symposium on Geotechnical Aspects of Underground Construction in Soft Ground, London, pp.713–718.

[28] Goh, K. H.; Mair, R. J. (2014) *Incorporating the existing condition of buildings when assessing potential impact of underground construction*. Soils and Foundations, No. 3, pp. 54.

[29] Burland, J. B.; Broms, B. B.; de Mello, V. F. B. (1977) *Behaviour of foundations and structures: State-of-the-art report*. Proceedings of 9th International Conference on Soil Mechanics and Foundation Engineering, Vol. 2, Tokyo, pp. 495–546.

[30] Boscardin, M. D.; Cording, J. C. (1989) *Building response to excavation-induced settlement*. ASCE Journal of Geotechnical Engineering 115, No. 1, pp. 1–21.

[31] Mair, R. J. (2013) *Tunnelling and deep excavations: Ground movements and their effects* in: Anagnostopoulos, A.E.A. (Ed.) Proceedings of the 15th European Conference on Soil Mechanics and Geotechnical Engineering – Geotechnics of Hard Soils – Weak Rocks (Part 4). IOS Press, pp. 39–70.

[32] Boon, C. W.; Ooi, L. H. (2016) *Tunnelling past critical structures in Kuala Lumpur: Insights from finite element analysis and T-Z load transfer analyses*. Geotechnical Engineering Journal of the SEAGS & AGSSEA Vol. 47, No. 4, pp. 109–122.

[33] Mair, R. J.; Williamson, M. G. (2014) *The influence of tunnelling and deep excavation on piled foundations. geotechnical aspects of underground construction in soft Ground* in: Yoo, Park, Kim & Ban (Eds) *Korean Geotechnical Society*, Seoul, Korea, ISBN 978-1-138-02700-8, pp. 21–30.

[34] Boon, C. W.; Ooi, L. H.; Tan, J. G.; Low, Y. Y. (2016) *Geotechnical considerations of deep excavation design for TBM bore-through with a case history in Kuala Lumpur*. Proceedings of GEOTEC, Hanoi, pp. 375–384.

[35] Khoo, C. M.; Gopalan, T.; Abdul Rahman, N. A.; Mohamad, H. (2019) *Volume loss caused by tunnelling in Kenny Hill formation*. International Journal of GEOMATE Vol. 16, No. 54, pp. 164-169. ISSN: 2186-2982 (Print), 2186–2990 (Online).

[36] Khoo, C. M.; Gopalan, T.; Abdul Rahman, N. A.; Terry, A. W. (2020) *Effects of EPB shield tunnelling on ground surface settlements – A case study in Kenny hill formation*. Proceedings of ITA-AITES World Tunnel Congress (WTC 2020), Kuala Lumpur.

[37] Mahalingam, T.; Khoo, C. M.; Chockalingam, P.; Cheng, S. S. (2015) *Design management and case study of underground construction adjacent to settlement sensitive building*. Proceedings of International Conference and Exhibition on Tunnelling and Underground Space (ICETUS 2015), Kuala Lumpur, pp. 280–284.

[38] Khoo, C. M.; Teoh, A. C. S.; Poh, S. T. (2019) *Building Responses due to Deep Excavation for Muzium Negara MRT Station, Kuala Lumpur*. Proceedings of ITA-AITES World Tunnel Congress (WTC 2019), Naples, Italy, pp. 5740–5747.

[39] Mak, W. K.; Cheah, F. L. P.; Ishak, A. (2015) *Challenges in Assessment of Existing Old Road Bridge due to Tunnelling Works*. Proceedings of International Conference and Exhibition on Tunnelling and Underground Space (ICETUS 2015), Kuala Lumpur, pp. 292–296.

[40] Khoo, C. M.; Teoh, A. C. S.; Terry, A. W.; Yau, K. F. (2020) *Challenges of tunnelling below the operating Pasar Seni LRT Station, Kuala Lumpur*. Proceedings of ITA-AITES World Tunnel Congress (WTC 2020), Kuala Lumpur.

[41] Vasagavijayan, S.; Ooi, C. E.; Shaiful, I. R.; Satkunaseelan, S. (2015) *Instrumentation and monitoring of underground works for KVMRT*. Proceedings of International Conference and Exhibition on Tunnelling and Underground Space (ICETUS 2015), Kuala Lumpur, pp. 220–229.

[42] Maxwell, A.; Vasagvijiyan, S.; Satkunaseelan, S. (2015) *The contribution of computer systems to the management of geotechnical and construction risk on the Klang Valley MRT SBK line underground works*. Proceedings of International Conference and Exhibition on Tunnelling and Underground Space (ICETUS 2015), Kuala Lumpur, pp. 286–290.

[43] Maxwell, A.; Chong, P. T.; Chin, M. Y. K. (2015) *Client-contractor collaboration for integrated TBM process and geotechnical control – power grid cable tunnels Singapore*. Proceedings of International Conference and Exhibition on Tunnelling and Underground Space (ICETUS 2015), Kuala Lumpur, pp. 106–110.

[44] Khoo, C. M.; Ooi, T. A.; Ng, S. W. F.; Feng, Q. Y. (2015) *An innovative approach of mined adit design and construction for removal of existing piles prior to TBM tunnelling*. Proceedings of International Conference and Exhibition on Tunnelling and Underground Space (ICETUS 2015), Kuala Lumpur, pp. 268–271.

[45] Tan, Y. C.; Teh, W. S.; Gue, C. S. (2019) *Special design considerations for underpinning system for existing structures due to tunnelling*. Proceedings of ITA-AITES World Tunnel Congress (WTC 2019), Naples, Italy.

[46] Ooi, T. A. (2015) *Tunnelling & underground space development in Malaysia. Feature article on variable density TBM (VD-TBM) and tunnelling development in Malaysia*. IEM Bulletin, Feb. 2015, pp. 6–9.

[47] Schaub, W.; Duhme, R. (2014) *Multi-mode and variable density TBMs latest trends in developments*. Underground Singapore 2014.

[48] Mohamad, H.; Tee, B. P.; Chong, M. F.; Lee, S. C. (2018) *Case studies of instrumented pile load tests with distributed optical fibre sensor as part of Klang Valley Mass Rapid Transit (KVMRT) project in limestone and granite formations*. IEM-CIE-HKIE Tripartite Seminar on Geotechnical Challenges in Infrastructure and Transportation Projects, Putrajaya, Malaysia.

[49] Chen, C. S. et al. (2019) *Measurement of shaft friction of bored pile in granite formation*. Proceedings of 1st Malaysian Geotechnical Society and Geotechnical Society of Singapore Conference, Petaling Jaya, Malaysia.

[50] Mohamad, H. et al. (2010) *Behaviour of an old masonry tunnel due to tunnelling-induced ground settlement*. Geotechnique 60, No. 12, pp. 927–938.

Authors



Ir. Chee-Min Khoo (corresponding author)
khoo.cheemin@mymrt.com.my
Mass Rapid Transit Corporation
5th Floor, Menara I&P 1
No. 46 Jalan Dungun, Bukit Damansara
50490 Kuala Lumpur
Malaysia



Ir. Dr Teik-Aun Ooi
drtaooi@gmail.com
TAO Consult Sdn Bhd
17A, Jalan Awan Hijau
Taman Overseas Union
Off Jalan Klang Lama
58200 Kuala Lumpur
Malaysia

How to Cite this Paper

Khoo, C. M.; Ooi, T. A. (2023) *Geotechnical challenges and innovations in urban underground construction – The Klang Valley Mass Rapid Transit project.* Geomechanics and Tunnelling 16, No. 3, pp. 243–262. https://doi.org/10.1002/geot.202300007

This paper has been peer reviewed. Submitted: 12.February 2023; accepted: 26. April 2023.

18657389, 2023, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/geot.202300007 by National University Of Singapore, Wiley Online Library on [18/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License